Using Meiss's action principle to quantify chaos in fusion devices

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Intérieur bourgeois anglais, avec des fauteuils anglais. Soirée anglaise. M. Smith, Anglais, dans son fauteuil et ses pantoufles anglais, fume sa pipe anglaise et lit un journal anglais, près d'un feu anglais. Il a des lunettes anglaises, une petite moustache grise, anglaise. A côté de lui, dans un autre fauteuil anglais, Mme Smith, Anglaise, raccommode des chaussettes anglaises. Un long moment de silence anglais. La pendule anglaise frappe dix-sept coups anglais.

Mme. SMITH

Tiens, il est neuf heures.

— Eugène Ionesco

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Abstract

In magnetically confined devices intended for fusion, it is common to witness sensitivity to initial conditions, i.e. chaotic behaviour. In the particular case of divertor design, it can be useful to trace field line and compute Poincaré/connection length plots. A map $\mathcal{P}$ that sends point from Poincaré section to the next is hereby defined. The chaos is then look at in term of the unstable/stable manifold tangle structure for saddle fixed point of $\mathcal{P}$. Following previous work from Meiss (2015) in the case of the Chirikov standard map, the turnstile flux, an important quantity to measure transport and the degree of chaos, is calculated in a Toy model. The method is then used to compute the turnstile fluxes in 3 configurations from the QUASR database and, finally, to look at the case of a chaotic configuration of W7X obtained, thanks to Robert Davies, by running current in the planar coils.

Contents

Introduction

While the Sun is essential to the Earth's energy supply, solar flares would tear away its atmosphere if there were no magnetosphere to act as a planetary shield. The magnetosphere is however not immutable, and substorms are disturbances that can cause the magnetic field in its tail, the region away from the Sun, to reconnect, creating a flow of plasma that returns to the Earth at the poles and can be seen in brightening of auroras. In fact, most of the known matter in the universe is in the plasma state, and its ionisation means that the charged particles follow the magnetic field, which is why the magnetosphere acts as a shield.

On Earth, a good understanding of the magnetic topology is essential to achieve a controlled fusion process in a reactor device. The electromagnetic fields determine to a great extent the evolution of the plasma, making it well or poorly confined. Although fusion can also be achieved by inertial confinement, which is currently the subject of research, it is not quite as mature as the magnetic confinement for fusion. Several different geometric designs have been explored, and the attempt to confine the plasma in a toroidal shape has proved most promising. A stability condition for good confinement is that the field must twist, causing the particles to move between low and high radii, effectively rotating around a particular field line called the magnetic axis.

This twisting is at the root of a schism in the design of toroidal magnetic fusion devices, leading to the two well known : Stellarator and Tokamak concepts. Both involve a magnetic field that is simultaneously toroidal and poloidal, rotating around the torus and around the magnetic axis. The toroidal component of the magnetic field is generated by the use of coils. They differ in the way the poloidal field is produced: Tokamaks induce a current in the plasma, which then generates the poloidal component according to Ampere's law. Stellarators have coils formed directly into modular shapes, eliminating the need for a net current, but are much more difficult to engineer and construct.

Start-ups have begun to emerge in the industry : Common Wealth Fusion (Tokamak), Renaissance Fusion (Stellarator), Type One Energy (Stellarator), to name a few. This is encouraging as competition within the industry might help explore and identify where the key failure points are. However, a number of challenges remain, including the design of the Plasma Facing Components (PCFs), in particular the design of a region that will bear most of the heat flux, the divertor. Under fusion-relevant conditions, it will be exposed "to a heat load that is ten times higher than that of a spacecraft re-entering Earth's atmosphere (10-20 MWm^2)" ITER. This is where some of the start-ups may miss the boat and try to move a little too fast, possibly making self-serving promises. Nevertheless, the link between industry and academia should be strong to ensure that progress in both areas is shared.

Many divertor concept are being investigated. In stellarator, a type of divertor exploits a resonance due to the winding of the field line to create magnetic islands at specific locations in the edge region, which direct the heat flux to certain spots in the plasma-facing components. Feng and W7-X-team (2022) reviews the divertor-relevant geometric parameters of the magnetic island divertor in the W7X stellarator.

Another type of divertor would be the non-resonant divertor (NRD). An NRD would be placed in trough regions : configuration-specific 'toroidal ridges' where the field lines primarily escape Bader et al. (2017). Garcia et al. (2023) study the edge structure for a possible NRD in the Compact Toroidal Hybrid device by looking at the connection length, the travelled distance before hitting a PFC numerically using the EMC3-EIRENE 3D code. They describe the striking patterns on a hypothetical divertor for different currents and at different divertor positions, and link it to chaos. They calculate the heat flux and Mach number and draw a parallel with the chaotic structure present in another type of divertor, the ergodic divertor. An NRD, unlike a resonant divertor, would not be subject to the position of a particular low order island which can be affected by the generated field/currents of the plasma motion itself. A NRD could therefore be a more robust design.

The stochasticity present in magnetically confined devices, described in great detail by Abdullaev (2014), leads to a third type of divertor concept, the ergodic divertor. Originally used to correct inaccuracies in toroidal coils that cause a non-axisymmetric field in tokamaks, resonant magnetic perturbation coils can also introduce mixing in the field line at the edge. It has been found to be beneficial for mitigating ELMs, a type of instability inherent to the high confinement regime in which, for example, the future DEMO reactor is designed to operate. The RMP coils are discussed as a means of creating an ergodic region in the edge, controlling the heat and particle fluxes, and directing the particles towards the divertor plates. The ergodic divertors have been used and studied in the WEST (formerly Tore Supra) and TEXTOR tokamaks.

The study of the edge plasma region is then essential for the design of a reliable divertors. As Imbert-Gerard et al. (2020) nicely points out, the complexity of describing the processes in a plasma can be tackled by considering different models depending on the length and time scale. Without taking into account the gyroscopic motion of the charged particles around the magnetic field and neglecting the interaction between the particles and the fields still provide a way to distinguish between different regions of the plasma in the stellarator. Following the movement of a point in the direction of $\mathbf{B}$ (forward) or $-\mathbf{B}$ (backward) gives is called field line tracing (FLT). It is usually used to draw the intersections or to view the connection length plot and provides significant amount of information about a configuration.

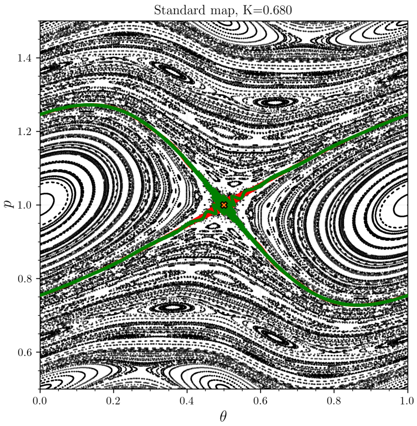
In the more general case of a 3 dimensional vector field, Wei and Liang (2023) describes the FLT in cylindrical coordinates as a map that shows the evolution of the crossing of a field line with a constant toroidal angle semi-plane. The resulting map, which I will call the field line or Poincaré map, is shown to have a determinant related to the divergence of the vector field. If the configuration is chaotic, then the trajectories depend strongly on the initial conditions, and two almost indistinguishably close initial conditions will eventually evolve into noticeably different regions of phase space. H. Poincaré was a precursor in this regard, looking at chaos in the case of celestial mechanics for the three-body problem. He associated chaotic behaviour with the appearance of an infinitely refined lattice structure, which he called a trellis in analogy with the construction of the interconnected garden fence (Poincaré, 1892, Ch.33). Nowadays it is also called a tangle. The trellis structure arises when looking at some specific points that are part of periodic orbits and is formed by the interconnection of two curves, the unstable and the stable manifold of one or more hyperbolic fixed point.

As Wei and Liang (2023) put it : "With the aid of modern dynamical system theory, the structure of a 3D vector field can be comprehended and analyzed in terms of invariant manifolds". In their paper they then show a hypothetical tangle in the case of an EAST shot when an RMP induced field would be added. In the case of field line tracing, we can analyse the 3-dimensional manifolds and tangle structure in a cross section with the field line map.

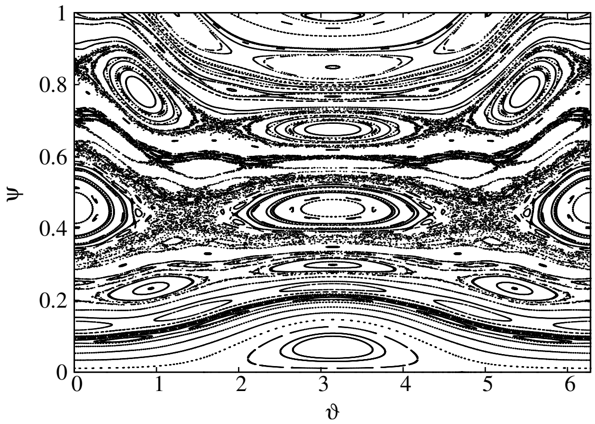
The field line map is continuous, but is a non-trivial problem: there is no general analytical formula, it can be lengthy to perform FLT, and the integration is prone to error. These difficulties make the study of the field line map not the best playground for understanding chaos. Simpler analytical maps are then a nice alternative and in particular the symmetric and asymmetric Tokamaps have been introduce to describe FLT in Tokamaks Abdullaev et al. (2006). There are based on a continous Hamiltonian system which consists in an integrable part specified by the winding profile, more precisely the safety factor q profile, and a non-integrable perturbation. Fig. 1b shows the orbits of the symmetric tokamap for a perturbed monotonic q profile. We can clearly see the difference between the region where the field lines form closed surfaces, the magnetic islands, and the regions where the points get mixed up, the chaotic zone. In fact, there may still remain some invariant tori in the chaotic sea. They are called cantori and their existence is proven by the KAM theorem.

Progress has been made in understanding chaos, for instance Easton (1991) began to describe transport. For volume-preserving[1] maps, Meiss (2015) provides the advances made. He looks at the well-studied Chirkov standard map, which the degree of chaos can be set through the parameter K. A phase portrait for $K = 0.68$ can be seen in Fig. 1a. In fact, the stable and unstable manifold curves are shown to bound a resonance region where the island lies, and for which there is an exit E and an entry I region. These exiting and entering sets are bounded between manifold intersections. The 'sizes' of the two sets are equal, there is as much flux coming from outside the resonance zone through the entry set as there is going out through

[1] Here the volume is a more general definition that can mean for instance the conservation of the magnetic flux in the case of the field line map.



(a)



(b)

Figure 1: Map for which the chaos is more easily studied and with similar behaviour as field line tracing : (a) standard map on the cylinder for the parameter $K = 0.68$. (b) symmetric tokamap taken from Wingen et al. (2005).

the exit set. By the similarity with a turnstile gates, the lobes are then called turnstiles. The size of the turnstile is more relevant for transport than another common quantity look at in chaos which is the Lyapunov exponent. Meiss puts it as :

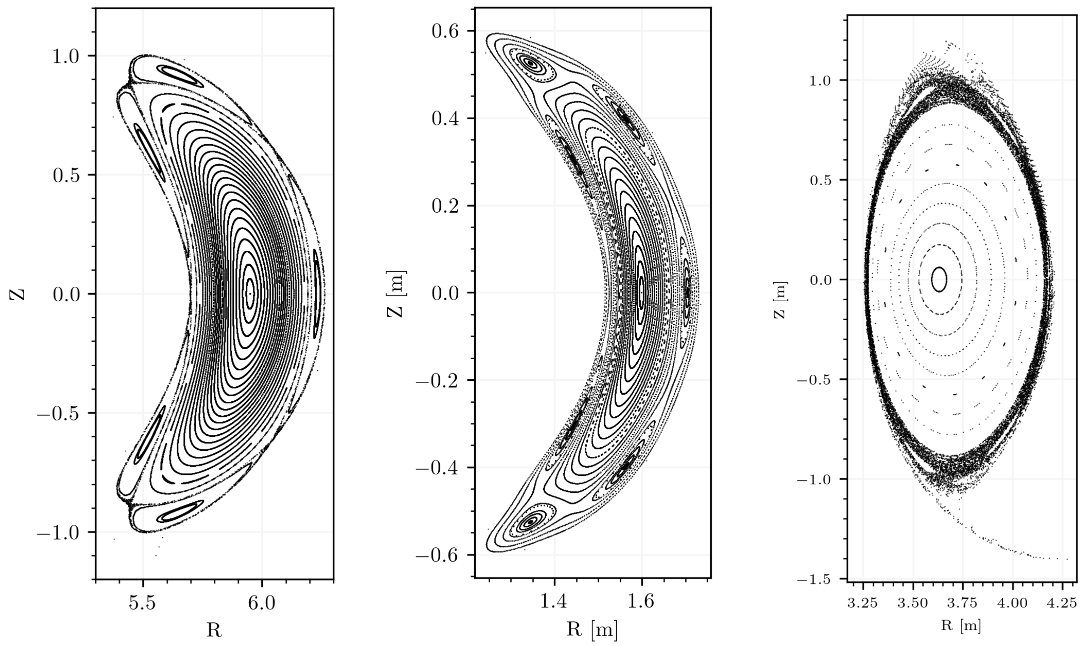
> "It is important to note that one of the fundamental measures of chaos, the Lyapunov exponent is not usually a good measure for transport. In Ref. 86, we said: The Lyapunov exponents give the rate at which a nice cat in a region of phase space turns into a mixed-up cat. The turnstile rate constants give the mean rate at which bits of the cat are transported to regions of the phase space"

Finally, it is worth noting that experimental signatures of turnstiles and tangles have been demonstrated Evans et al. (2005). According to the authors' calculations, the pattern of chaotic magnetic field lines hits on the vessel is related to the arrangement of the perturbed system tangles and experiments in which heat and particle distributions were measured on the PFC sometimes exhibited split peaks, indicating the presence of chaotic fields and homoclinic tangles.

In the following paper, we use Hamiltonian dynamical systems theory, in particular the turnstile flux/area which is a measure of chaotic transport, to analyse magnetic configurations in fusion devices. We develop a simplified analytical model for toroidal fields with arbitrary perturbations Ch.2 that is capable of reproducing islands and fields with a tokamak-like divertor. We then use this model to demonstrate a fast, stable algorithm for calculating turnstile area in Ch.3, Sec.3.3.1 based on an action principle introduce by Meiss in Meiss (2015). Finally we apply this method directly to quantify the chaos in stellarator fields supplied directly from coils. We analyze three different QUASR configurations and one chaotic current configuration of W7X. Finally we show preliminary results that indicate that the turnstile area, and hence transport on the outboard side of the island is larger than the inboard, and explain how features in connection length plots are related to the turnstiles.

1 Field line map

In the study of continuous dynamical systems, it is often convenient to consider a discrete map defined by the intersection of the trajectories with a lower-dimensional subspace and which captures some properties of the continuous case. For particles following the magnetic field of a toroidal apparatus, the intersection of their trajectories with a constant ϕ cross-section reveals the distinct region of different field line behaviour : closed surfaces, islands, chaotic regions, and is known as a *Poincaré* section.



(a) W7-X, $n_{\mathrm{fp}} = 5$

(b) NCSX, $n_{\mathrm{fp}} = 3$

(c) LHD, $n_{\mathrm{fp}} = 10$

Figure 1.1: Poincaré section at $\phi = 0$ for (a) Wendelstein-7X, (b) the (aborted) National Compact Stellarator Experiment and (c) the Large Helical Device stellarator.

Using Biot-Savart law to get the field generated from the known currents and coil geometry and by neglecting the gyro-motion around the field lines, the crossings can be numerically recorded. In Fig. 1.1 the Poincaré sections for three well known devices are computed using Runge-Kutta integration scheme. For W7X and NCSX configuration, we see that there is one smaller island chain which is predominant. For LHD on the contrary, the edge region is chaotic and multiple small islands are present in the chaotic edge region. Such a figure can be produced experimentally in stellarators by firing an electron beam, sweeping a fluorescent rod and using a long camera exposure Pedersen et al. (2016). To define a map that gives the evolution of the position of the crossings in the ϕ semi-plane, first note that for each initial point $\mathbf{x}_0$ there is an associated curve $\gamma(t)$ which is link to the flow $\gamma(t) = \Phi(\mathbf{x}_0, t)$ of the magnetic vector field. The cylindrical coordinates are a natural choice here.

Geometric consideration : Throughout this document, the convention adpoted is that of the contravariant component for vectors. That is, for example, $\mathbf{B}$ in cylindrical coordinates has the form $B^R = \mathbf{B}\cdot\nabla R$, $B^\phi = \mathbf{B}\cdot\nabla\phi$ and $B^Z = \mathbf{B}\cdot\nabla Z$. In the formalism of differential geometry, one writes $B = B^i\partial_i$. The basis $\{\partial_i\}_i$ is orthogonal and holonomic but the element may not be unitary, as for ∂_ϕ. On the other hand, the component in the common orthonormal basis e_i will be written with subscript as $\tilde{B}^i$. See App. A. for more details.

The trajectory curve $\gamma : \mathbb{R}^+ \to \mathbb{R}^3 =: \mathcal{M}$, which is $t \mapsto \gamma(t) = (R(t), \phi(t), Z(t))$ in cylindrical coordinates we are interested in γ such that the tangent vector at each point is given by $\dot{\gamma}(t) = \mathbf{B} = B^i\partial_i \in T_{\gamma(t)}\mathcal{M}$. Specifying the initial point $\gamma(0) = \mathbf{x}_0$ identifies a specific curve :

$$\gamma(t) = \int_0^t \dot{\gamma}(s)ds + \mathbf{x}_0 \quad \text{with} \quad \dot{\gamma}^i(t) = (\frac{dR}{dt}, \frac{d\phi}{dt}, \frac{dZ}{dt}) = (B^R, B^\phi, B^Z).$$

Provided that the sign of $B^\phi(R,\phi,Z)$ is either positive or negative on γ, B^ϕ is never zero, the map :

$$t(\phi) = \int \frac{dt}{d\phi} d\phi = \int_0^\phi \frac{1}{B^\phi} d\phi$$

defines a handy change of variable for which the ϕ evolution is linear. Indeed γ can be re-parameterized as $\phi \mapsto \gamma(\phi) = (R(\phi), \phi, Z(\phi))$ with $\dot{\gamma}$:

$$\dot{\gamma}^i(\phi) = (\frac{dR}{d\phi}, 1, \frac{dZ}{d\phi}) = (\frac{dR}{dt}\frac{dt}{d\phi}, 1, \frac{dZ}{dt}\frac{dt}{d\phi}) = (B^R/B^\phi, 1, B^Z/B^\phi).$$

Due to the choice of coordinates the field will have at most a ϕ periodicity of 2π, yet in stellarator configurations have azymuthal redundancy. Their period is $T = 2\pi/n_{\text{fp}}$ where $n_{\text{fp}} \in \mathbb{N}^\star$ is the number of field period in a complete toroidal rotation. For example, W7X and

LHD have 5 and 10 field periods respectively. If the field is not periodic then $n_{\text{fp}} = 1$ and, in contrast, for an axisymmetric device $n_{\text{fp}} = +\infty$, the case of ideal Tokamaks.

The Poincaré section is identical for ϕ_i and $\phi_i + kT$, $k \in \mathbb{Z}$. Writing Ω the set of initial points in the ϕ_i plane for which γ is effectively re-parametrizable between ϕ_i and $\phi_i + T$, allows to define the map $\mathcal{P} : \Omega \to \mathbb{R}_+ \times \mathbb{R}$ as :

$$(R, Z) \mapsto \mathcal{P}(R, Z) = \int_{\phi_i}^{\phi_i+T} (\dot{\gamma}^R(s), \dot{\gamma}^Z(s))\, ds + (R, Z) \tag{1.1}$$

with $\mathbf{x}_0 = (R, \phi_i, Z)$. The point $(R, Z) \in \Omega$ are point in the initial section and should not be confused with the γ^R and γ^Z components. Fig. 1.2 represent an hypothetical $n_{\text{fp}} = 3$ configuration where an initial starting point $x_0 := (R_i, Z_i)$ gets map to the $\phi_i + T$ cross section. The evolution of the intersection for x_0 is then simply $x_{i+1} = \mathcal{P}(x_i)$ and $x_n = \mathcal{P}^n(x_0)$.

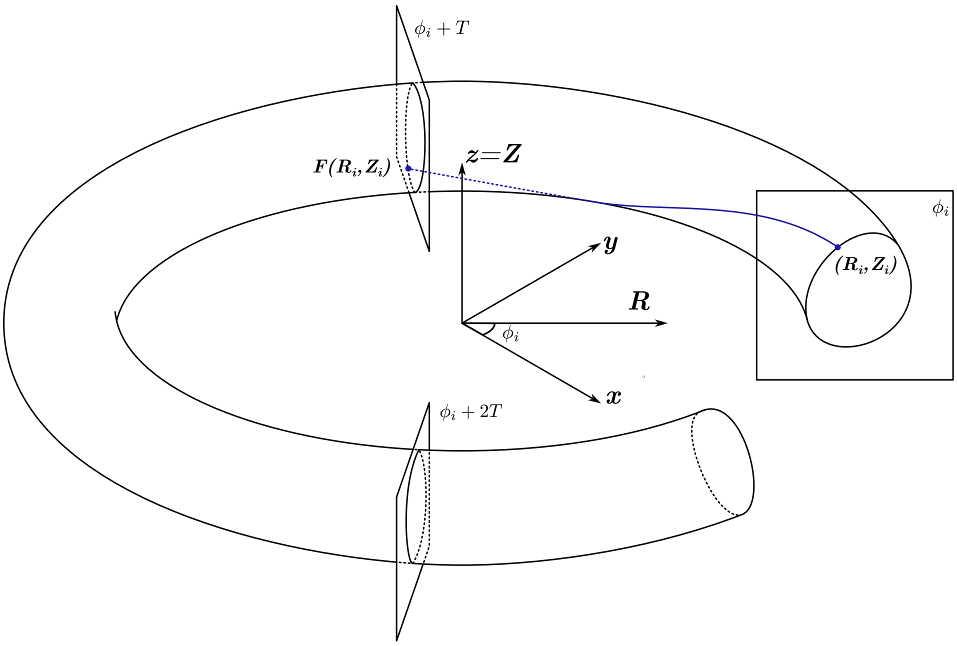


Figure 1.2: Trajectory of a particle in a toroidal device (Tokamak/Stellarator) from a starting point (R, Z) in a $\phi = \phi_i$ section to a point in the $\phi = \phi_i + T$ plane with T the azymuthal periodicity of the magnetic field.

As the evolution is performed by following $\mathbf{B}$ and due to the magnetic field been divergence free $\nabla \cdot \mathbf{B} = 0$, the flux through any surface obtained by mapping a simple path in ϕ_i to $\phi_i + T$ will be zero. This result in the key property of $\mathcal{P}$ being flux-conserving ; the flux through any closed surface $\Sigma \subset \Omega$ is equal to the one through $\mathcal{P}(\Sigma)$:

$$\iint_{\Sigma} \mathbf{B} \cdot \mathbf{dS} = \iint_{\mathcal{P}(\Sigma)} \mathbf{B} \cdot \mathbf{dS}.$$

The determinant of the Jacobian matrix for a general flow $\Phi(\mathbf{x}_0, t)$ of a vector field v, which does not depend on time, is related to the divergence of v (Hirsch et al., 2013, p.408) by :

$$\det(\mathcal{D}\Phi(\mathbf{x}_0, t)) = \det(\mathcal{D}\Phi(\mathbf{x}_0, 0)) \exp\left(\int_0^t \mathrm{Tr}(\nabla v(s))\, ds\right) = \exp\left(\int_0^t \nabla \cdot v(s)\, ds\right) \tag{1.2}$$

with $v(\Phi(\mathbf{x}_0, s))$ written implicitly as $v(s)$. In the case of the flow in cylindrical coordinates, Wei and Liang use Eq. 1.2 to derive the relation :

$$\det(\mathcal{D}\Phi(\mathbf{x}_0, \phi_i, \phi_e)) = \exp\left(\int_{\phi_i}^{\phi_e} \frac{R(\nabla \cdot v)}{\tilde{v}^\phi}\, ds\right) \frac{\tilde{v}^\phi(\phi_i)}{\tilde{v}^\phi(\phi_e)}.$$

It gives for the determinant of $\mathcal{D}\mathcal{P} \in \mathbb{R}^{2\times 2}$ that :

$$\det(\mathcal{D}\mathcal{P}(R, Z)) = B^\phi(R, \phi_i, Z) / B^\phi(\mathcal{P}^R(R, Z), \phi_i + T, \mathcal{P}^Z(R, Z)). \tag{1.3}$$

The last identity may also be recovered more directly using differential forms as in Meiss (2015), the derivation is given in App. A.2.

1.1 Fixed points

The field line map $\mathcal{P}$ given by Eq. 1.1 defines a map from Ω to Ω. Such a map has interesting properties. Take for example Brouwers's fixed point theorem, e.g. Michel Chipot (2008), which states that any map from the disk to the disk has at least one fixed point. This means that there must be at least one point, which through the field line map, ends up exactly where it started! In fusion this point, and the field line through it has a special name, it is called the 'magnetic axis' (and is often used as the 'axis' of a coordinate system constructed on the basis of a magnetic field). More generally, a point $x^\star \in \Omega$ mapped to itself after k iteration of the field line map is part of a periodic orbits, representing the crossing of a closed field line.

The magnetic field twist around the magnetic axis in order to average out the $\nabla \mathbf{B}$ drift. The twisting is quantify by the rotational transform $\iota\!\!\!-$ which indicates the fraction of poloidal turns performed after one toroidal turn. When $\iota\!\!\!- = n/m$ is rational, each field line rotates an angle $2\pi\iota\!\!\!-$. That means that each field line comes back to where it started after at most m complete transits around the torus. For such a surface, it can be energetically more favorable to break up into an 'island chain'; instead of every field line returning to the same surface, there remains one closed field line around which the other field lines wind around, called the O-point of the island, and another field line with winding number n/m of that surface called the X-point.

In Tokamaks, islands break the axisymmetry of the field and reduce confinement. They are then unwanted. The action by which they are created is called a 'tearing mode', as the unbroken surfaces are 'torn open'. In Stellarators, there is no underlying symmetry and the islands can be a present, static feature of the equilibrium. Sometimes, like in W7X they are even desired and optimized for, as they generate a divertor-like structure Feng and W7-X-team (2022). Due to its relation with magneto-hydrodynamic stability limit, it is also more common to use the safety factor $q = 1/\iota\!\!\!-$ in Tokamaks.

X/O-points of a $\iota\!\!\!- = n/m$ magnetic island close after m toroidal turns. As there is n_{fp} symmetry planes in one turn, they are a fixed point of $\mathcal{P}^{m \cdot n_{\text{fp}}}$. But the minimum number of iterations to close the orbit $k \in \mathbb{N}^\star$ is called the order and is not in general equal to $m \cdot n_{\text{fp}}$. After k application of the field line map the winding around the axis is $\Delta\theta(k) = 2\pi\iota\!\!\!-\, k/n_{\text{fp}} = 2\pi \cdot n/m \cdot k/n_{\text{fp}}$. To make one full rotation k must then be $k = m\, n_{\text{fp}}/n$, which is the correct order for of the orbit. The number of island is the greatest common divider between m and n. NCSX stellarator is with 3 field periods and an island in the edge with $\iota\!\!\!- = 3/7$, shown in Fig. 1.1b. There is only $1 = \gcd(3,7)$ island that rotates around the axis and the number of application of $\mathcal{P}$ for one of its fixed point to loop around is $k = 7 \cdot 3/3 = 7$.

To distinguish between O and X points, the behaviour of nearby field lines can be described using the Taylor expansion :

$$\mathcal{P}^k(x^\star + \delta x) = x^\star + \langle \mathcal{D}\mathcal{P}^k|_{x^\star}, \delta x\rangle + \mathcal{O}(\|\delta x\|^2).$$

Especially by looking at the eigenvalues λ_1 and λ_2 of the jacobian $\mathcal{D}\mathcal{P}^k|_{x^\star}$. Using Eq. 1.3, the determinant of $\mathcal{D}\mathcal{P}^k|_{x^\star}$ is equal to 1 or equivalently the matrix is an element of $\mathrm{SL}(2,\mathbb{R})$. Depending on the value of the trace the eigenvalues are either :

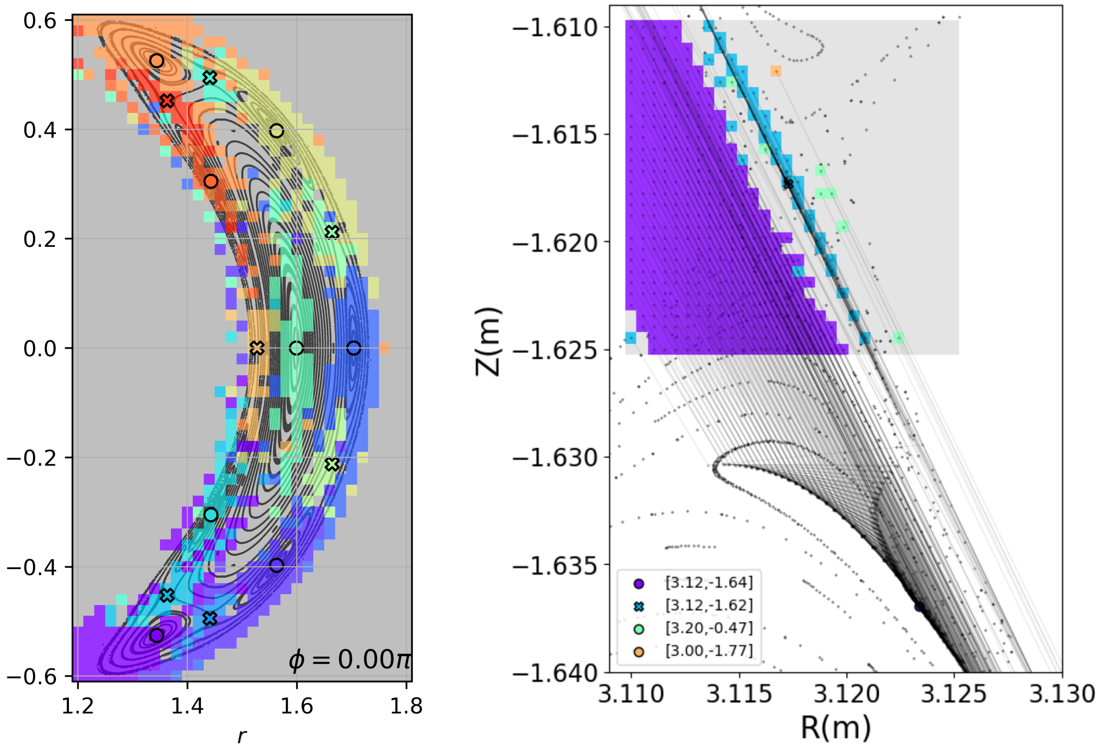
1. Complex conjugate and of modulus 1 ; $\lambda_{1,2} = e^{\pm i\theta}$, when $|\mathrm{Tr}(\mathcal{D}\mathcal{P}^k|_{x^\star})| < 2$. They indicate rotation around the fixed point making it an O-point.

2. Both real with $|\lambda_s| < 1$ and $|\lambda_u| > 1$, when $|\mathrm{Tr}(\mathcal{D}\mathcal{P}^k|_{x^\star})| > 2$. When they are positive, it indicates an X-point.

3. $\lambda_1 = \lambda_2 = \pm 1$.

To find the fixed point of the field line map the Newton method will be used, see App. B.3. The O-points are relatively easy to find when using an initial guess in the right region. However, for many X-points, the strong exponential behaviour around them make the jacobian matrix ill conditioned. Fig. 1.3 shows the region of convergence around the X/O-point of NCSX and of a Toybox Tokamak, as the one presented in next section but with smaller aspect ratio.

For NCSX we see that the region of convergence, for example around the two upper X-points, is small compared to that of the nearby O-points. Even more, we can see that there are points close to the X-point at $Z = 0$ that converge to the O-point on the other side of the configuration. For the Toy Tokamak, the region of convergence is even smaller. It converges quickly to another O-point of a $m = 1$ island. The step taken in the Newton method are heavily directed along a diagonal, which was found to coinciding with the position of starting point that have a ill condition Jacobian matrix.



(a) (b)

Figure 1.3: Convergence region of the Newton algorithm for (a) NCSX configuration for the X and O points of the $n/m = 3/7$ island. (b) Analytic single null Toy-Tokamak. The point in (R,Z) are coloured by the point where they converged to, marked as X/O. The point that are not converging are coloured in grey.

The Poincaré or field line map $\mathcal{P}$ was defined as the map which takes the intersection of a field line with a given poloidal section and moves it to its next intersection. By considering the periodicity of the field, it was shown that it was not necessary to integrate a complete toroidal rotation. The flux conservation of the field line map was discussed and its relation with the determinant of the Jacobian was shown. The relationship between the fixed point of the $\mathcal{P}$ and the X/O point of the magnetic islands was investigated. Finally the sensitivity of the Newton method to the initial condition and the poor convergence region for the hyperbolic points was shown.

2 Toybox

An analytical magnetic geometry is introduced to provide a simpler and more flexible approach to the field line tracing analysis. This *Toybox* is designed in cylindrical coordinates. Its baseline is the field of a high aspect ratio Tokamak equilibrium with quadratic q-profile, that can be tuned. On top of which the field generated by a circular current loop and perturbation can be added/removed and modified. It will be simpler than treating the geometry of a complex toroidal device and be handy to validate the numerical calculation of further algorithms. From the linearity of the curl operator, a sum of vector potentials $\mathbf{A}_i$ has a one-to-one correspondence with the element of the resulting magnetic fields $\mathbf{B}_i$. Defining the model in this way has the advantage of making it possible to use automatic differentiation from the *JAX* package Bradbury et al. (2018). The vector potential is the only quantity that needs to be defined analytically. The resulting $\mathbf{B}$ or, in fact, any derivative of any reasonable order of A can then be computed. The creation of a tokamak-like single null equilibrium is then the main focus.

More specifically in axisymmetric equilibria (i.e. perfect Tokamaks), following e.g (Wesson, 2011, p.108), the shape of the magnetic surfaces are well defined by the flux function ψ. Indeed, for cylindrical coordinates, all components/functions are constant along ϕ, which yields :

$$\mathbf{B} = B^i \partial_i = -\frac{1}{R}\frac{\partial \psi}{\partial Z}\partial_R + \frac{1}{R}\frac{\partial \psi}{\partial R}\partial_Z + \left(\frac{\partial A^R}{\partial Z} - \frac{\partial A^Z}{\partial R}\right)\partial_\phi.$$

with $\psi = R^2 A^\phi$. The choice of Gauge can be exploited to assume $A^Z = 0$. For completeness, note that in the formulation of the Grad-Sharfranov equation, the last term, giving the ϕ component of the field, is related to a function $F(\psi)$ as $(\mu_0)F/R^2$.

A convenient geometry to study for the flux surfaces is nested toroids centered on the magnetic axis of RZ-coordinates (R_0, Z_0). Introducing toroidal coordinates (ρ, ϕ, θ), with ρ the distance from the axis, the poloidal flux is thus $\psi(R, Z) \propto (R - R_0)^2 + (Z - Z_0)^2 = \rho^2$. In fact, due to the freedom in the dimension of the implementation, the equality can be choosen. For large aspect ratio tokamaks $R_0/\rho >> 1$ the safety factor profile can be approximated by :

$$q \approx \frac{\Delta\phi}{\Delta\theta} = \frac{\rho\tilde{B}^\phi}{R_0\tilde{B}^\theta} = \frac{2\pi}{\mu_0\tilde{I}^\phi R_0}\rho^2$$

calculating the multipole expansion of $\tilde{B}^\theta$ only up to first order. The $\tilde{B}^\phi$ component and the plasma current $\tilde{I}^\phi$ are imposed by the toroidal field coils and the central solenoid respectively. It would be desirable to achieve this quadratic behaviour of q in the tokamak model as well. In more details, the q-profile can be calculated in a single ϕ section by integrating over the poloidal closed curve defined by $\psi = \text{const}$ (Wesson, 2011, pp.111-112). In this specific case, as $\psi = \rho^2$, this path simply reduce to an integration over θ as :

$$q(\rho) = \frac{1}{2\pi}\int_0^{2\pi}\frac{d\phi}{d\theta}d\theta = \frac{1}{2\pi}\int_0^{2\pi}\frac{B^\phi}{B^\theta}d\theta. \tag{2.1}$$

The θ component can be derived directly from coordinate transformation, as follows :

$$B^\theta = \frac{\cos\theta}{\rho}B^Z - \frac{\sin\theta}{\rho}B^R = \frac{2}{R} = \frac{2}{R_0 + \rho\cos\theta}.$$

The ϕ component of $\mathbf{B}$[1] is introduced heuristically as :

$$B^\phi = 2\sqrt{R^2 - \rho^2}\,(q_a + \frac{\alpha_s}{2}\rho^2)/R^2 \tag{2.2}$$

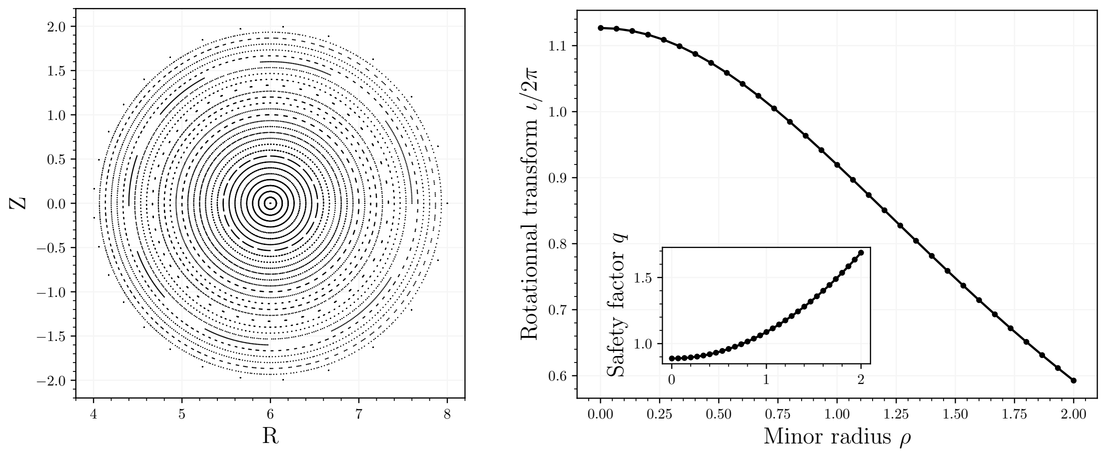
with its corresponding A^R provided in App. B.1. Inserting B^θ and B^ϕ in (2.1) gives :

$$q(\rho) = (q_a + \frac{\alpha_s}{2}\rho^2)\frac{\sqrt{R^2-\rho^2}}{2\pi}\int_0^{2\pi}\frac{1}{R_0 + \rho\cos\theta}d\theta = q_a + \frac{\alpha_s}{2}\rho^2$$

where the last two term cancel for $\rho < R$. At this point, we can make sense of the and α_s constants as the on axis safety factor and the actual shear $dq/d\rho$. Fig. 2.1a shows the cross section obtained by tracing the field line over a full toroidal revolution for the equilibrium with $R_0 = 6$, $Z_0 = 0$, $q_a = 0.8875$ and $\alpha_s = 0.4$. While the effective number of field periods is $n_{\text{fp}} = \infty$, calculating the Poincaré section by integrating over 2π remains correct even when perturbations are added and the axisymmetry breaks. The density of intersections is higher on the inside than on the outside. Since $B^\rho = 0$ and $\tilde{B}^\phi/\tilde{B}^\theta \propto \sqrt{(\rho/R)^2 - 1}$ for constant ρ, a field line revolves faster in ϕ when $\theta \sim \pi$ then when $\theta \sim 0$ and thus make more crossings.

The rotational transform $\iota\!\!\!-$ of field lines are computed by integrating both the field line themselves and the magnetic axis, tracking the evolution of the winding. Performing such a calculation for starting points of the form $(R_0 + \rho, 0)$ with $\rho \in [0,2]$ gives the $\iota\!\!\!-$ profile in Fig. 2.1b.

[1] In the implementation used in this document, the B^ϕ has been oriented along the -∂_ϕ direction, which has no direct effect on the calculation themselves, but may occasionally cause some sign reversal, e.g. the latter value of the current in the circular loop coil. It also flips the way the field line rotate around the axis.



(a) (b)

Figure 2.1: High aspect ratio Toy-Tokamak equilibrium with $R_0 = 6$, $Z_0 = 0$, $q_a = 0.8875$ and $\alpha_s = 0.4$. (a) Poincaré section computed with Runge-Kutta algorithm. (b) Rotational transform $\iota/2\pi$ and q profile with respect to the distance ρ from the axis.

The calculated q-profile is in great agreement with its analytical form, which is reassuring.

An essential feature of the diverted tokamak edge structure is the separatrix. Combining a quadratic q equilibrium with the field generated by a circular current loop at appropriate position and amplitude will result in a single null equilibrium. The vector potential of a curent loop at position $(R_l, 0)$ is given in Simpson et al. (2001) as :

$$A^{\phi}_{\text{sep}} = \frac{\mu_0}{4\pi}\frac{4IR_l}{\beta R}\left(\frac{(2-k^2)K(k^2)-2E(k^2)}{k^2}\right).$$

with K, E the complete elliptic integral of the first and second kind and with $\alpha^2 = (R_l - R)^2 + Z^2$, $\beta^2 = (R_l + R)^2 + Z^2$ and $k = 1 - \alpha^2/\beta^2$. To get a general position from this expression, Z can be substituted with $Z - Z_l$.

The superposition of $\psi_{\text{sep}} = R^2 A^{\phi}_{\text{sep}}$ with $I = 10\pi/\mu_0$, $R_l = 6$ and $Z_l = -5.5$ on the ψ_{eq} for nested toroids with $R_0 = 6, Z_0 = 0, q_a = 0.91, \alpha_s = 1.2$ is shown in Fig. 2.2a. ψ_{sep} produces only a field lying in the poloidal plane and with the separatrix coil current parallel to the plasma current, the poloidal component of the field cancels out somewhere between the former axis and the position of the loop, producing the desired X-point. The resulting analytical Tokamak equilibrium is shown in Fig. 2.2c. The separatrix appears clearly. It can be appreciated that for small distances with respect to the axis the q-profile is still quadratic Fig. 2.2c and tends to infinity as it approaches the separatrix.

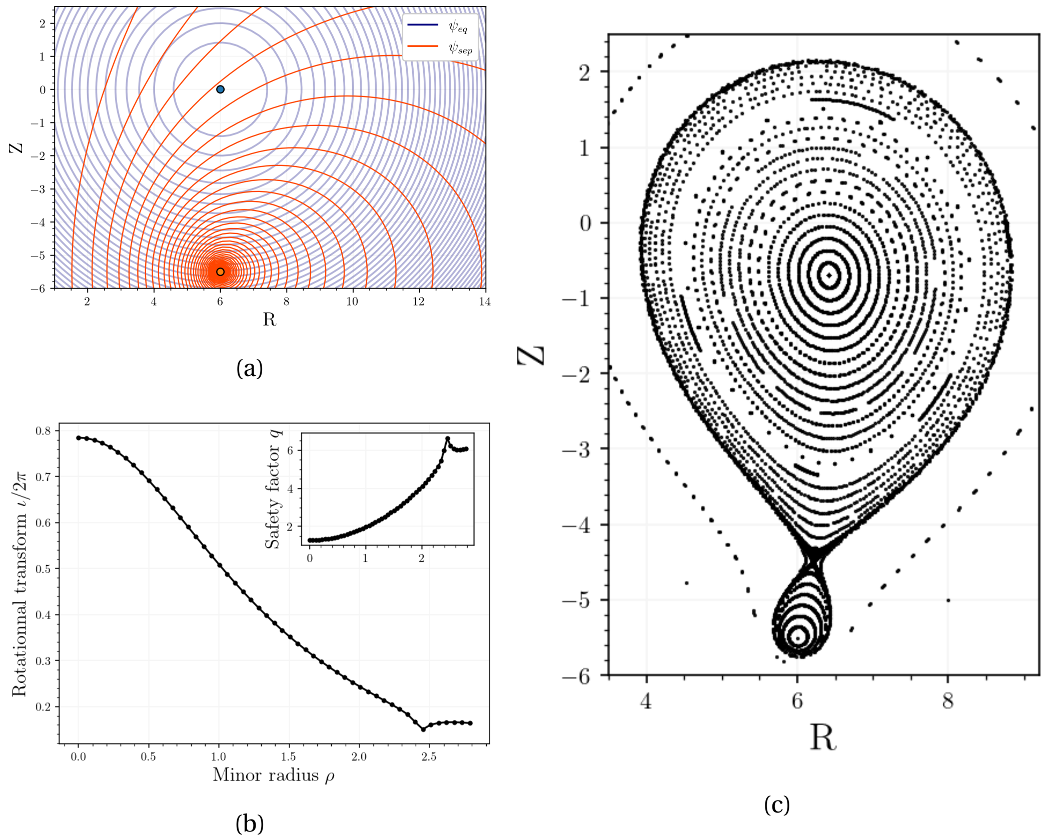


Figure 2.2: Analytical Tokamak equilibrium obtained by (a) the addition of the ψ_{sep} function generated by a current loop on top of a high aspect-ratio toroids equilibrium. (b) $\iota/2\pi$ and q profiles and (c) resulting Poincaré section.

2.1 Perturbations

Since the equilibria generated in the *Toybox* so far are axisymmetric, Noether's theorem implies that there can be only closed flux surfaces and no chaos. To make it appear, perturbations $\delta\mathbf{A}$ must be applied. Using simple toroidal coordinates ϕ, θ and $0 < \rho < R_{\text{max}}(\theta)$, makes any function periodic in θ and ϕ. Thus one can develop any component of the perturbation into a Fourier series as :

$$\delta A^i(\rho,\phi,\theta) = \sum_{n\in\mathbb{N}}\sum_{m\in\mathbb{N}} \delta A^i(\rho)\cos(n\phi+\varphi_n)\cos(m\theta+\varphi_m)$$

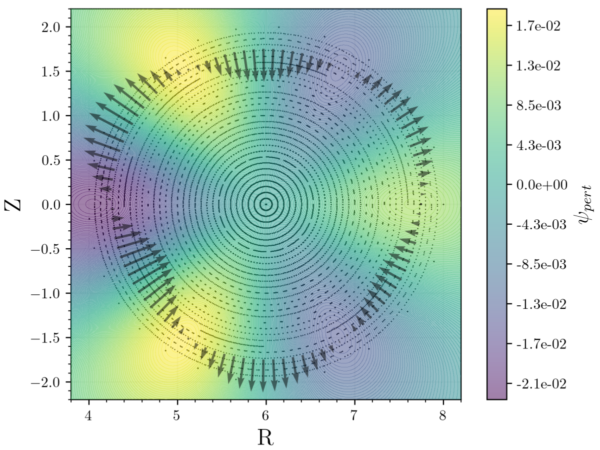
Escande and Momo (2024) note that while $\delta\mathbf{A}$ generally has 3 components, with a suitable redefinition of the perturbation in straight field line coordinates $\delta\mathbf{A}$ can always be of the form $\delta\mathbf{A} = -\delta\psi\nabla\phi$ with only a ϕ component. Although ρ and θ are generally not straight field line coordinates, using the same approach, writing $\psi_{\text{pert}} = R^2\delta A^\phi$ and taking only one Fourier mode pair (m,n) will provide the building block for the toybox perturbations.

The resonant surface has a specific q value and can be targeted, but applying the perturbation to a more widespread region was thought to be more effective and somewhat realistic. For the radial profile of ψ_{pert}, $f(\rho) = R^2\delta A^{\phi}(\rho)$, the Normal and Maxwell-Boltzmann distribution are implemented :

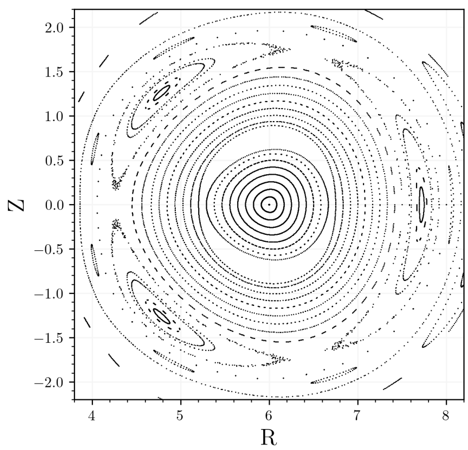
1. A Normal distribution $f(\rho,\mu,\sigma) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(\frac{-(\rho-\mu)^2}{2\sigma^2}\right)$ with mean and mode μ and variance σ^2.
2. A Maxwell-Boltzmann distribution $f(\rho,d) = \frac{\sqrt{2}}{\sqrt{\pi}}\frac{\rho^2}{d^3}\exp\left(\frac{-\rho^2}{2d^2}\right)$ with mode $\sqrt{2}d$.

The last parameter that can be select in this implementation is the amplitude $\varepsilon_{\mathrm{amp}}$ of the linear combination $\mathbf{A} + \varepsilon_{\mathrm{amp}}\delta\mathbf{A}$. A fair way to compare the perturbations, for example by their magnetic energy, could be investigated, but is currently lacking.

First we apply a Maxwell-Boltzmann distributed perturbation to the high aspect ratio equilibrium with the mode pair $m = 3$, $n = 2$ and to target the $q = 3/2$ surface we set $d = \rho/\sqrt{2}$ and $\varepsilon_{\mathrm{amp}} = 0.1$. The resulting perturbation flux ψ_{pert} is shown in Fig. 2.3a. The more yellow the more ψ_{pert} is positive and the more purple the more negative. The perturbed magnetic field at $\rho = 1.75$ is shown, it follows the curves of the constant ψ_{pert}. For the divergence-free condition to be satisfied, the flux/field intensity must be higher on the inside, which is indeed observed. Fig. 2.3b shows the resulting poincare section obtained. The crossings of the $q = 3/2$ island are clearly visible. There also seems to be one island closer to the axis and one outside.



(a)



(b)

Figure 2.3: High aspect ratio Tokamak equilibrium perturbed with Maxwell-Boltzmann distributed perturbation $m/n = 3/2$, $d = 1.75/\sqrt{2}$ and $\varepsilon_{\mathrm{amp}} = 0.1$. In (a) the contour of the perturbed flux ψ_{pert} and the resulting perturbed field on the $\rho = 1.75$ circle. In (b) Poincaré section at $\phi = 0$ for the perturbed magnetic configuration and the 3 crossing of the island.

While the resulting perturbed high aspect configuration does not appear chaotic at first, we will further see that there is indeed chaos here. In fact the modes of perturbation are not only the m and n which are used in the implementation because the toroidal coordinates are not straight field line coordinates. Fig. 2.4 shows the angle between the $B^\phi \partial_\phi$ and $\mathbf{B}$ obtained by the $\arctan(B^\theta / B^\phi)$ with respect to the angle position θ, ϕ on the $\rho = 1.75$ surface in the case of the unperturbed high aspect ratio equilibrium. The highest values are more red and indicate that the field is more oriented towards ∂_ϕ. Lower values are shown in blue. The angle changes with θ, which is not the case for straight field line coordinates[2], where it would be constant and equal to $\iota\!\!\!-(\rho) = 2/3$. This distinction introduces mode mixing, the initial delta distribution in Fourier space is broadened and thus chaos appears faster, which is a welcome feature in this case.

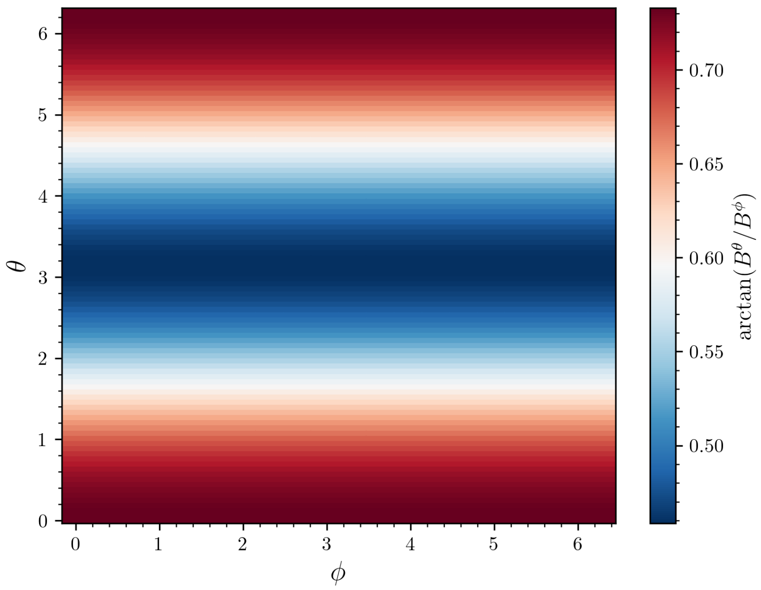


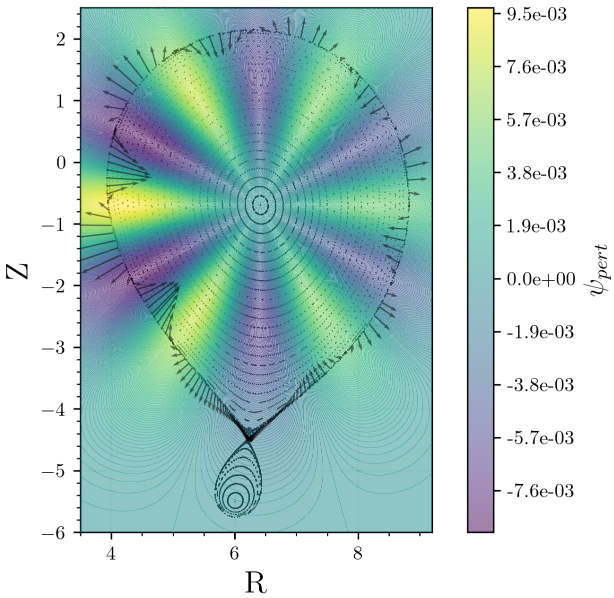
Figure 2.4: Angle between $B^\phi \partial_\phi$ and $\mathbf{B}$ in radian against the poloidal θ and toroidal ϕ angles for the magnetic surface at $\rho = 1.75$ and with $q(\rho) = 3/2$.

A Maxwell-Boltzmann distributed perturbation can also be added to the single null Tokamak equilibrium. Checking the safety factor profile Fig. 2.2b suggest picking a pair of modes $m = 6$, $n = 1$ and $d = \sqrt{2}$ (to target $\rho = 2$) and $\varepsilon_{\text{amp}} = 0.1$. Similar to the high aspect ratio case, the Fig. 2.5a shows the ψ_{pert} over the single null tokamak equilibrium. The orientation changes from pointing inward to outward 12 times. The Poincaré section of the resulting field is shown in figure Fig. 2.5b.

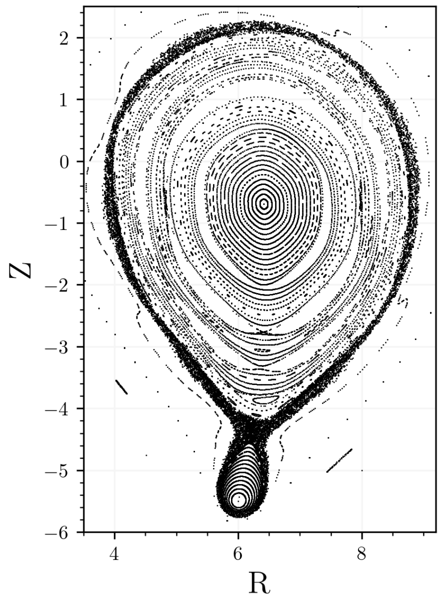
The regime can be here seen to be chaotic and there are many islands/island chains. A set of points starting close to the X-point make a more dense layer on the position of the previous separatrix. The position of the O-point is not affected by the perturbation and the X-point is shifted by a relative change of order 10^{-3}. A closer look at the field lines around the X-point is shown in Fig. 2.6 with different colours for each field line and the X-point as a crossed

[2]The basis transformation can be found for example in (D'haeseleer et al., 1991, p.116).

marker. It reveals how the field lines are mix in this region. Some higher order islands are also discovered.



(a)



(b)

Figure 2.5: Single null Tokamak equilibrium perturbed with Maxwell-Boltzmann distributed perturbation $m/n = 6/1$, $d = \sqrt{2}$ and $\varepsilon_{\text{amp}} = 0.1$. In (a) the contour of the perturbed flux ψ_{pert} and the resulting perturbed field on the unperturbed separatrix. In (b) Poincaré section at $\phi = 0$ for the perturbed magnetic configuration.

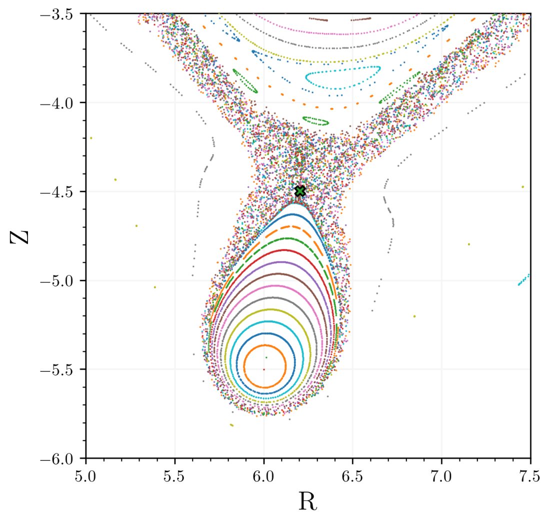


Figure 2.6: Zoom in view around the X-point of the single null Tokamak equilibrium showing the chaotic behaviour of the nearby field line.

In this chapter we have presented an analytical model for the magnetic field. We used it to calculate a high apsect ratio and single null tokamak-like equilibrium. We then introduced a magnetic perturbation in toroidal coordinates for the magnetic field and showed that chaos arises, and discussed the effect of non-straight field line coordinates.

3 Tangles and Turnstiles

Hamiltonian dynamical systems can show a certain dependence on the initial condition. Such that starting points that are almost equal at time t_0 evolve to completely different results, while the system is deterministic![1] This has often been described as the butterfly effect, where a flap of the wind from a butterfly on the other side of the world is enough to change the initial state so that the end result - let's not wish for bad things - a snowstorm occurs or not. However, this seemingly erratic behaviour is structured by the appearance of ever-repeating patterns : fractals.

H. Poincaré, in his work on celestial mechanics Poincaré (1899), was considering asymptotic solution of the three body motion problem and remarked an infinite intersection behaviour of the asymptotic trajectories. The two curves were forming a infinitly fine grid, which he called a trellis because of its resemblance to the strange garden wall. The trellis, now also referred to as tangle, is key in understanding the chaos. Hohloch and Hohloch (2017) provides a rigorous mathematical definitions for those concepts.

In the case of volume-preserving maps, Meiss (2015) relates the asymptotic curves, called in fact unstable and stable manifold, to a measure of transport. Indeed he defines a resonance region for which there are gates to go in and out, the exiting and incoming fluxes, which forms a turnstile. He points out that the characterisation of transport in deterministic dynamical systems involves the computation of exit and transition time distributions in phase space.

In the case of magnetic confinement, the curve described by following the field line describes a Hamiltonian system and is prone to chaos. Viana et al. (2023) states that under certain conditions the equations for magnetic field lines can be reformulated in canonical form, also argued by Escande and Momo (2024). In this paradigm, Abdullaev (2014) proposes two mapping methods to study magnetic field lines near the separatrix of poloidal divertor tokamaks in the presence of external non-axisymmetric magnetic perturbations. They use them for the study of the properties of open stochastic field lines near the separatrix.

[1] A really nice book to read about is Chaos by James Gleick.

For the field line map, the study of tangles and turnstiles requires some numerical algorithms : computing the manifolds, finding the homo/heteroclinics, calculating the turnstile fluxes are all tasks that need to be tackled. The algorithms presented here will be part of the *Pyoculus* package and will hopefully be useful for a better understanding of chaos in the future.

3.1 Stable and Unstable Manifolds

The exponential behaviour along the stable and unstable directions, given by the stable $\mathbf{e}_s$ and unstable $\mathbf{e}_u$ eigenvectors, of a saddle fixed point $x^\star$ can be expressed by stating that for small ε the points $x_s = x^\star + \varepsilon\,\mathbf{e}_s$ will converge to $x^\star$ as we apply $\mathcal{P}$ and the points $x_u = x^\star + \varepsilon\,\mathbf{e}_u$ converge to $x^\star$ as we apply $\mathcal{P}^{-1}$. More specifically, the stable manifold $W^s(x_s^\star)$ is defined as the set for which $\mathcal{P}^k(x) \to x_s^\star$ as $k \to \infty$. Similarly, the unstable manifold $W^u(x_u^\star)$ is the set of points such that $\mathcal{P}^k(x) \to x_u^\star$ as $k \to -\infty$. These definitions includes not only the points in the linear regime, but also the points forming their trajectory. In the following, the manifolds are simply written as W^s and W^u both when $x_s^\star = x_u^\star$ and when it is not.

In the case of a two dimensional map, such as for the field line map, Easton (1986) states that the stable manifold theorem proves that there exist immersions γ_s and γ_u mapping $\mathbb{R}$ to W^s and W^u with $\gamma_s(0) = \gamma_u(0) = x^\star$. It implies that the point on the stable and unstable manifold can be ordered by induction of the order of $\mathbb{R}$, yielding the stable and unstable orderings $>_s$ and $>_u$. Intuitively, this means that if we take the fixed point $x^\star$, there are two curves that we can follow : the stable and unstable manifold and two direction for each corresponding to the orientations $\pm\mathbf{e}_s$, $\pm\mathbf{e}_u$. It gives then four components W_+^s, W_-^s, W_+^u and W_-^u which corresponds as well to the mapping of $\mathbb{R}^+$, $\mathbb{R}^-$ through γ_s and γ_u.

Each of the four components, for example W_+^s, can then be described by picking $q \in W_+^s$ and considering the segment $S^+ =]q, \mathcal{P}(q)]_s \subset W_+^s$. By applying $\mathcal{P}$ and $\mathcal{P}^{-1}$ to S^+ for a sufficient number of iterations, the segment will be going forward or backward and draw the manifold. It is then possible to recover the whole manifold just by considering S^+.

For a single null tokamak equilibrium, the separatrix consists of two components, for instance $W_+^s = W_+^u$. They form in principle a 3-dimensional surface. Since their evolution is continuous in ϕ, their 2-dimensional intersection with a phi-plane is considered. In the chaotic case, W_+^s and W_-^u are no longer equal. A simple algorithm to draw/compute them then consists to follow the evolution of two segments S^+ and $\mathcal{U}^+$.

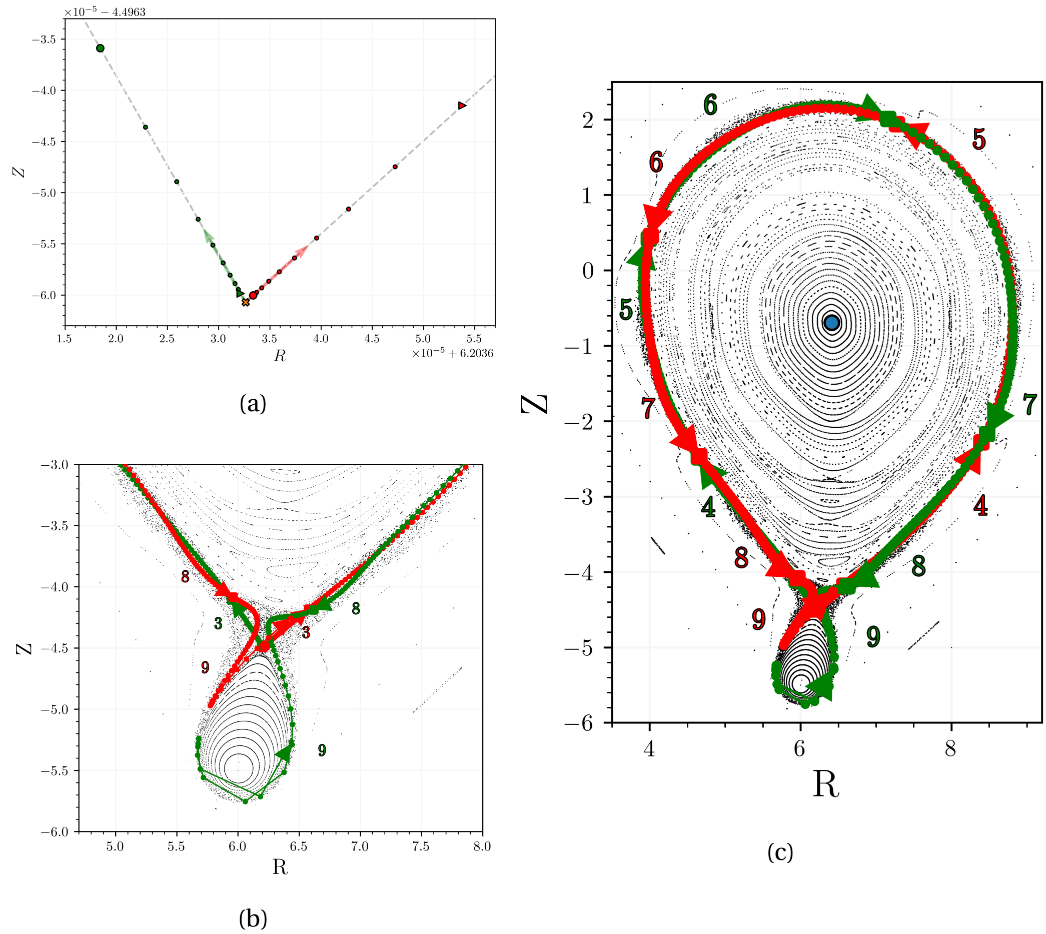


Figure 3.1: Different steps of the manifold plotting process for the perturbed Toy-Tokamak. The starting points set in the linear regime are shown in (a) while (b) and (c) shows the evolution of the segments $\mathcal{U}^+$ and S^+ as the field lines are traced forward/backward.

The algorithm used, proposed in Wei and Liang (2023), is illustrated in Fig. 3.1. It consists in initialising a point on each of the manifolds: $x_s \in W^s_+$, $x_u \in W^u_+$ well in the linear regime and getting the segments $S =]x_s, \mathcal{P}^{-1}(x_s)]_s$ and $\mathcal{U} =]x_u, \mathcal{P}(x_u)]_u$ by tracing the point for one crossing. Then as $\mathcal{S}$ and $\mathcal{U}$ are still in the linear regime they can be discretized with $x_{i,s/u} = x^\star + \varepsilon_i\, \mathbf{e}_{s,u}$. And the initial ε_i can be set to be equally spaced in the exponential space e^{λ_s} and e^{λ_u} to better capture the change in behaviour.

Fig. 3.1a shows how applying $\mathcal{P}^\pm$ on the starting point denoted by circle markers : $x_{i,s}$ in green and $x_{i,u}$ in red, map them respectively closer and further away to the point marked with a triangle. By further mapping the $\mathcal{S}$ and $\mathcal{U}$ segments with the backward and forward evolution, the entire manifolds are recovered. Fig. 3.1b illustrates the progress as the number of iterations increases. The initially short length of the segments increase as they leave the linear regime. At a certain point they become slightly shorter, but as they approach the saddle again they get increasingly stretched out. Fig. 3.1c shows how they start to wrap around the separatrix coil and go back and forth as they continue to expand. We see how this creates the tangle.

3.2 Hetero/Homo-clinic points

The intersection of W^s and W^u are of particular relevance as they are points for which $\mathcal{P}^k(x)$ asymptotically converges to $x_s^\star$ for $k \to \infty$ and $x_u^\star$ for $k \to -\infty$. The points p in $\mathcal{H} = W^s \cap W^u / \{x_s^\star, x_u^\star\}$ are called heteroclinic points if $x_s^\star \neq x_s^\star$ and homoclinics otherwise. One immediate thing to note is that if at some point we have h a homoclinic point. Then any evolution point of h forward or backward will still be a homoclinic point. An equivalence class for the points p, q in $\mathcal{H}$ is given by : $[p] = [q]$ if p can be obtained by mapping the point q for some number of forward or backward iterations. The 3d trajectory of a homo/heteroclinic in the toroidal device is linked to the 2d crossings by the relation $[p_i] = [p_j]$ if they belong to the same field line.

A particular type of intersection, the *primary* homo/hetero-clinic points, is of interest. They are defined formally in (Hohloch, 2017, p.14) and they represent geometrically the intersections $p \in \mathcal{H}$ between the two manifold for which the curves that joins $x^\star$ to p in W^s and W^u only intersect at their endpoint. An algorithm for finding these crosspoints is presented here, based on an approach suggested by Matt Landermann. In Sec.3.1, it has been seen that after 6 forward/backward iterations the two curves $\mathcal{P}^{-6}(\mathcal{S})$ and $\mathcal{P}^{6}(\mathcal{U})$ are intersecting. Writing $x_{0,s}$ and $x_{0,u}$ two starting points in $\mathcal{S}$ and $\mathcal{U}$, the goal is to find the root of :

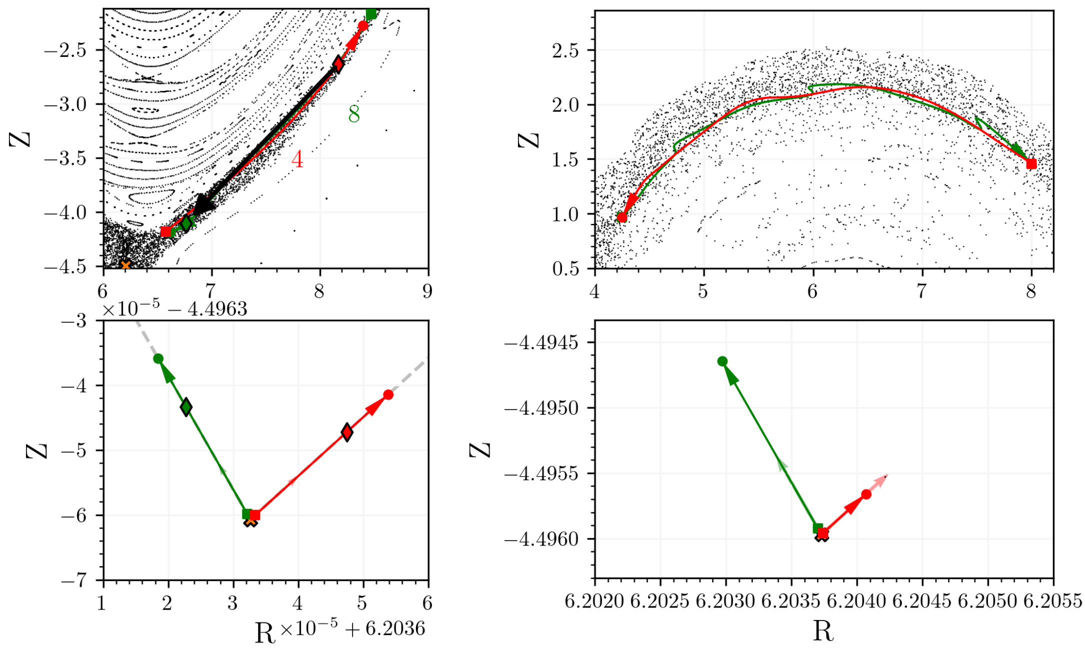
$$\mathcal{P}^{-6}(x_{0,s}) - \mathcal{P}^{6}(x_{0,u})$$

which can be achieved by an optimization process[2]. In a more general context, if $\mathcal{S}$ and $\mathcal{U}$ cross after $-j$ iteration for $\mathcal{S}$ and k iteration for $\mathcal{U}$, then every (n_s, n_u) pair such that their sum equals $j+k$ will also give images of the segments that intersect. The hetero/homoclinics can be searched in a region closer to the fixed point, where the shape of one of the manifolds is a bit more twisted, making the algorithm converge better. Fig. 3.2a shows the initial step for the primary homoclinic search in the perturbed Toy-Tokamak we have been looking at. The initial crossing $6+6$ was shifted to $n_s = 8$ and $n_u = 4$. The optimisation tries to find starting points such that the length of the black arrow is zero.

This works well to find a first homoclinic point. However there are at least another primary homoclinics to find. The initial points for the segment $\mathcal{S}$ and $\mathcal{U}$ can be retaken with ε_s^1 and ε_u^1 the values for the first homoclinic point found. Now, as Fig. 3.2b shows it, the segments will cross but have equal starting and ending points. The number of intersection in between are then equal to the number of other primary homoclinics that are yet to be found.

In the single null Toy-Tokamak with 18/3 perturbation, there are a total of 6 primary crossings. And in the case of the 12/2, there was found to be 4 primary homoclinics. It seems that the relation is simply the mode n, but some other perturbations examined did not follow this

[2]In this case and for all the configuration discussed, scipy.optimize.root default algorithm was used.



(a)

(b)

pattern. The number of primary homoclinics being greater than two may be due to the mixed mode perturbation, it is hard to say for sure. The question is really interesting and it may well be that there is some clever way of relating them with the mode perturbation to the mathematics of prime numbers. This needs to be explored further [3]. The numbers were found to always come in pairs in the toy box, but no explanation was found as to why.

Here a clever way to search for the remaining intersection of the two curve may exist. However using the same idea as for the first one, minimizing this time $\mathcal{P}^{-n_s-1}(x_s) - \mathcal{P}^{n_u}(x_u)$, but using a starting point of the form :

$$x_s = x^\star + \lambda_s^{i/n_h}\varepsilon_s^1 \mathbf{e}_s \quad x_u = x^\star + \lambda_u^{i/n_h}\varepsilon_u^1 \mathbf{e}_u \quad \text{with} \quad i = 1, \ldots, n_h$$

where n_h is the number of homoclinics to find. In most cases the procedure converge to the right homoclinic. This intuitively works because a starting epsilon will be mapped to $\lambda_{s,u}\varepsilon$. Then by taking points of the form $\lambda_{s,u}^{i/n_h}$, we will have n_h points with equal separation in exponential space $\exp(\lambda_{s,u})$.

In general the optimization problem tackled here is not an easy one. The distance between the end points have plateau when the two segments are travelled in opposite directions. This has an analogy for instance with the Rosenbruck function optimization problem, where the low gradient in a wide valley give the algorithm a hard time. A robust understanding and way of knowing the number of primary homoclinics and an appropriate method are the welcome

[3] Perhaps there is already some literature on the subject.

and exciting next steps.

3.3 Turnstile

In non chaotic case, magnetic islands are bounded by a last close flux surface but when chaos is present, the separatrix is broken into the stable and unstable manifold. Using the definition of primary homo/heteroclinic points, a region bounded by the manifolds can still be defined. It is such that the interior contains the O-point. In the heteroclinic case, there are two pair of stable/unstable manifold that define the boundary by taking the initial segment $[x_s^\star, p] \cup [p, x_u^\star]$ and $[x_s^\star, q] \cup [q, x_u^\star]$ with p and q the primary heteroclinic, one on each pair of W^s, W^u. Meiss calls this region the resonance zone Meiss (2015). He looks at the (0,1) orbit of the well studied Chirikov standard map on the cylinder :

$$p_{n+1} = p_n - \frac{K}{2\pi}\sin(2\pi\theta_n),$$
$$\theta_{n+1} \equiv \theta_n + p_{n+1} \pmod 1,$$

when $K = 1.5$. The K parameter, which plays the role of a perturbation, is a convenient way to introduce chaos. As it increases, so does the chaos. Fig. 3.3a shows the resonance region for the (0,1)-orbit and highlighting the stable and unstable manifolds. By taking the initial segments of m_1^+ for the upper and m_1^- for the lower homoclinics, the resonance zone is recovered. W^s and W^u are also the boundaries of smaller lobes that we can see at the edge of the resonance zone. The dark green lobe for example. Points in one lobe are mapped to another through the standard map, shown by the f arrows. It can then be realized that this behaviour will lead a set on the interior of the resonance region to be mapped out, precisely here the dark green set. But also for a set of point on the outside to be mapped in the region. Due to the exactness property of the standard map, the sum of the area of exiting sets E is equal to that of the entry sets I.

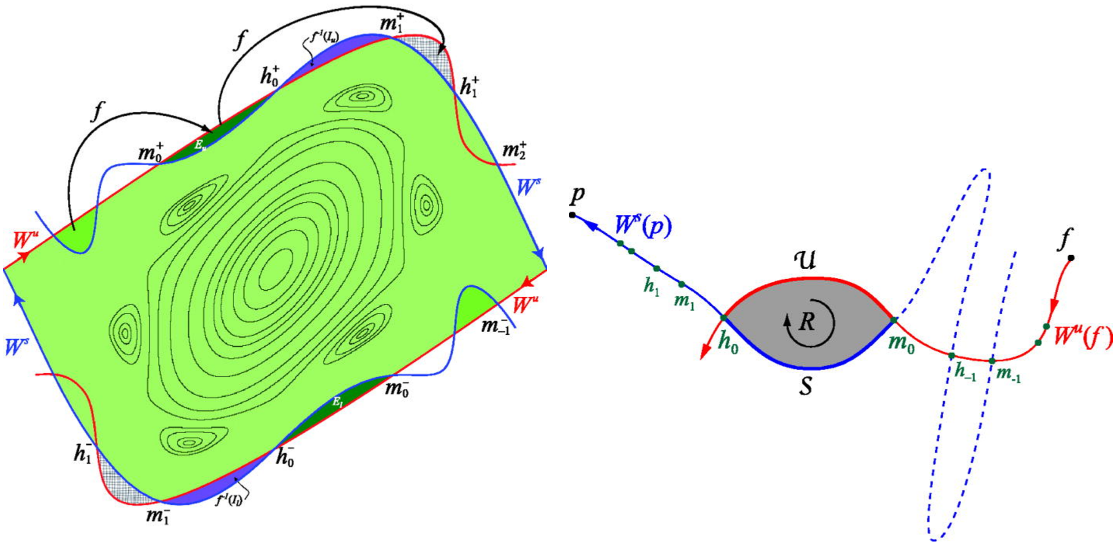
The area of the lobes indicates the degree of chaos. The lobe will be stretched out in a folding and squeezing process, but its area will remain the same. Meiss introduces a formula for how to compute this indicative area in a general map that preserves an exact volume form $\Omega = d\alpha$ and is exact :

$$\mu(R) = \int_R \Omega = \int_{\partial R} d\alpha = \int_{\mathcal{U}} \alpha - \int_{\mathcal{S}} \alpha = \sum_{t=-\infty}^{\infty} (\lambda(m_t) - \lambda(h_t)) \tag{3.1}$$

with λ a Lagrangian-like function found from α and f. The path of the integral are given in Fig. 3.3b. This obscure mathematical formulation can be clarified slightly for the standard map ; as $1 = \nabla \times (x\mathbf{e}_y)$ we can use Stokes' theorem to write :

$$\mu(R) = \int_R 1 = \int_{\partial R} \nabla \times (x\mathbf{e}_y) = \int_{\mathcal{U}} x\mathbf{e}_y \cdot \mathbf{dl} - \int_{\mathcal{S}} x\mathbf{e}_y \cdot \mathbf{dl}. \tag{3.2}$$

Then the last equality uses exactness and surprising link between $\mathcal{U}$ and $\mathcal{S}$ that $\mathcal{U} = f^n(\mathcal{S})$ to write the sum on the homoclinics intersections m_i^+ and h_i^+ for instance.



(a) (b)

Figure 3.3: Schematic representation of a turnstile in (a) for the resonance zone of the (0,1)-orbit of the standard map with $K = 1.5$, (b) for the tangle structure formed by the manifolds $W^s(p)$ and $W^u(f)$ of two f and p saddles. In both cases the orbit intersections of primary homo/heteroclinics are denoted by h_i and m_i. The figures are taken from Meiss (2015).

3.3.1 Adapting Meiss's action principle for the map $\mathcal{P}$

Looking at the field line map, there is a nice analogy with the conservation of flux and B equals $\nabla \times \mathbf{A}$. More precisely looking at the mathematical description of Maxwell's equations using differential forms. In the case of $\mathcal{P}$ we don't know a priori that there is only one lobe for the turnstile, we do expect however for the total sum $\mu(E) = \mu(I)$. Calculating the flux through one lobe is still the most general thing to do.

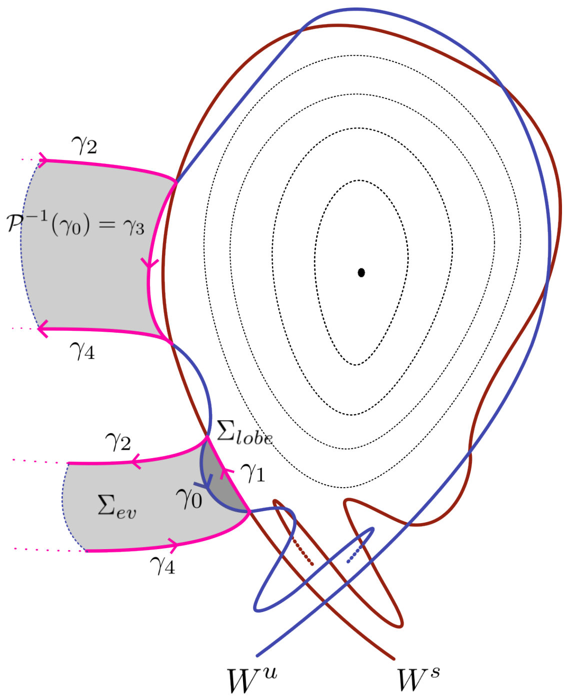


Figure 3.4: Schematic representation of the calculation of the turnstile area in a homoclinic tangle. The tangle is formed by W^s and W^u in a chaotic configuration of the Toy-Tokamak with perturbation $m/n = 6/1$. A turnstile lobe is depicted as Σ_{lobe} with boundary $\gamma_0 + \gamma_1$ and the boundary of a more complex surface with identical flux $\Sigma = \Sigma_{lobe} + \Sigma_{ev}$ is especially emphasised in pink.

The boundary $\partial\Sigma_{lobe}$ of a specific lobe Σ_{lobe} of the turnstile can be subdivided, as illustrated in Fig. 3.4, into two curves : $\gamma_0 \subset W^u$ and $\gamma_1 \subset W^s$. γ_0 joins the intersections h_0 of the homoclinic h field line to the intersection m_0 for the second homoclinic m and γ_1 returns from m_0 to h_0. Using Stokes theorem, the flux Φ_{lobe} through Σ_{lobe} can therefore be written as :

$$\Phi_{lobe} = \int_{\Sigma_{lobe}} \mathbf{B} \cdot \mathbf{dS} = \int_{\Sigma_{lobe}} \nabla \times \mathbf{A} \cdot \mathbf{dS} = \int_{\gamma_0+\gamma_1} \mathbf{A} \cdot \mathbf{dl}. \tag{3.3}$$

To directly, numerically, calculate the integral in Eq. 3.3 requires the curves γ_0 and γ_1 to be discretized. In practise, it would mean that many initial positions need to be set in the linear regime of W^u and W^s in between position of h and m, then their field line have to be traced up until they form the boundary $\partial\Sigma_{lobe}$. The number of points must be sufficient and appropriately tuned to describe the bending of the manifolds well enough to avoid introducing too many errors. Although it is not necessary to locate the homoclinics beforehand, and further approximations (such as assuming that the points describe a Bezier curve) are possible improvements to such a method, it becomes very expensive to increase the accuracy as it means more integrations have to be performed. This will prove impractical in a general stellarator field. A more elegant algorithm for calculating the lobe flux is suggested, based on the principle of the turnstile calculation for the standard map presented by Meiss.

Tracing the curve γ_0 backward around the torus generates a surface Σ_{ev}, depicted in Fig. 3.4, for which the normal $\mathbf{dS}$ is always orthogonal to the magnetic field $\mathbf{B}$ according to definition of FLT. It means that the flux through $\Sigma := \Sigma_{lobe} + \Sigma_{ev}$ is equal to the one through Σ_{lobe} and Eq. 3.3 becomes :

$$\Phi_{lobe} = \int_{\Sigma_{lobe}} \mathbf{B} \cdot \mathbf{dS} = \int_{\Sigma_{lobe}+\Sigma_{ev}} \mathbf{B} \cdot \mathbf{dS} = \int_{\partial\Sigma} \mathbf{A} \cdot \mathbf{dl}. \tag{3.4}$$

The oriented boundary $\partial\Sigma = \gamma_1 + \gamma_2 + \gamma_3 + \gamma_4$ is composed of four paths : $-\gamma_2$ and γ_4 are obtained as the trajectories of the homoclinics h and m from $h_{-1} := \mathcal{P}^{-1}(h_0)$ and $m_{-1} := \mathcal{P}^{-1}(m_0)$ to h_0, m_0 around the torus. γ_1 and γ_3 which will be referred to as joining curves and lie in the Poincaré section. As for γ_0, γ_3 is part of the unstable manifold and while γ_0 connects h_0 to m_0, γ_3 joins h_{-1} to m_{-1}. A flat representation of the tangle structure is shown in Fig. 3.5. The unstable manifold, initially straight around the fixed point $x_u^\star$, has an increasing curvature as it approaches $x_s^\star$. The stable manifold shows a similar behaviour with the points reversed. Their intersections h_i and m_i are shown as a circle and a square respectively, and the joining integral path can be seen more clearly.

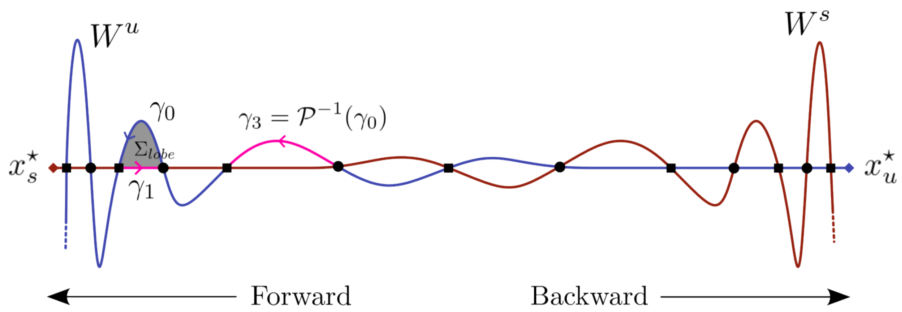


Figure 3.5: Flatten out version of the manifold tangle structure between two saddle fixed points $x_s^\star$ and $x_u^\star$. The forward evolution $\mathcal{P}$ maps points close to $x_u^\star$ in the direction of $x_s^\star$ and the backward evolution the opposite. Two different hetero/homo-clinics are present and are denoted by the squared and circular markers. The surface of the lobe Σ_{lobe}, as well as the joining curves γ_1 and γ_3 are highlighted.

With respect to the integration path taken in Eq. 3.3, the path of (3.4) contains the trajectory of the homoclinic as part of the contour. The relationship with the formula given by Meiss appears, however, the difficulty of finding γ_1 and γ_3 remains.

To take care of those joining integrals, note that the argument for Σ_{ev} is still true by applying the map more than once backward on γ_0. Indeed, by repetitively applying the backward evolution, $\mathcal{P}^{-k}(\gamma_0) \subset W^u$ joins h_{-k} and m_{-k} that are closer and closer as they approach $x_u^\star$. The length of the the curve converges to 0 as $t \to \infty$ and using Cauchy-Schwartz inequality, the integral of $\mathbf{A} \cdot \mathbf{dl}$ also converges to zero :

$$\int_{\mathcal{P}^{-k}(\gamma_0)} \mathbf{A} \cdot \mathbf{dl} \le \int_{\mathcal{P}^{-k}(\gamma_0)} \|\mathbf{A}\| \, \|\mathbf{dl}\| \le L(\mathcal{P}^{-k}(\gamma_0)) \max_{\mathbf{x} \in \mathcal{P}^{-k}(\gamma_0)} \|\mathbf{A(x)}\| \lesssim L(\mathcal{P}^{-k}(\gamma_0)).$$

Following the trajectory of the homoclinic points h_0 and m_0 until they converge to the fixed point, the length of γ_1 reduces to zero and it's contribution to the integral also converges to zero. Since the flux through Σ_{lobe} and $\mathcal{P}^k(\Sigma_{lobe})$ is equal (again, flux conservation of $\mathcal{P}$), the same idea as for γ_3 can be applied to γ_1. Applying the map repetitively will make the length of $\mathcal{P}^k(\gamma_1)$ and thus its contribution to the flux calculation tend to zero. Including these considerations in Eq. 3.4 and taking the the number of forward and backward iterations to infinity gives :

$$\Phi_{lobe} = \lim_{k\to\infty} \int_{m_{-k}}^{m_k} \mathbf{A}\cdot\mathbf{dl} - \int_{h_{-k}}^{h_k} \mathbf{A}\cdot\mathbf{dl} \tag{3.5}$$

where the similarity with the equation provided for the standard map (Meiss, 2015, Eq. 21) is now striking. The complicated description of the joining curves, that would require hundreds of point are reduce to the calculation of $\mathbf{A}\cdot\mathbf{dl}$ on only two trajectories!

The calculation of Eq. 3.5 requires enough iterations for the distance between the homoclinics to converges to 0. However, due to the exponential behaviour around the X-points and the inherent lack of infinity in double precision, once the field line is close to the fixed point, only a small number of crossings will be recorded before the numerical error causes the point shifted away, ending the possibility of convergence. To overcome this, and in fact to make the algorithm much more robust, a small tweak is introduced. In Eq. 3.3 the difficulty of a numerical computation would have been to get the shape of the manifold precisely. Here instead for a certain finite value of k : $\mathcal{P}^k(\gamma_2)$ and $\mathcal{P}^{-k}(\gamma_0)$ are mapped back into the linear regime of $x_s^\star$ and $x_u^\star$. There, the manifolds are simply segments and the joining integrals can be approximated by the midpoint rule, which requires only few evaluations of the vector potential $\mathbf{A}$, but no additional field line tracing.

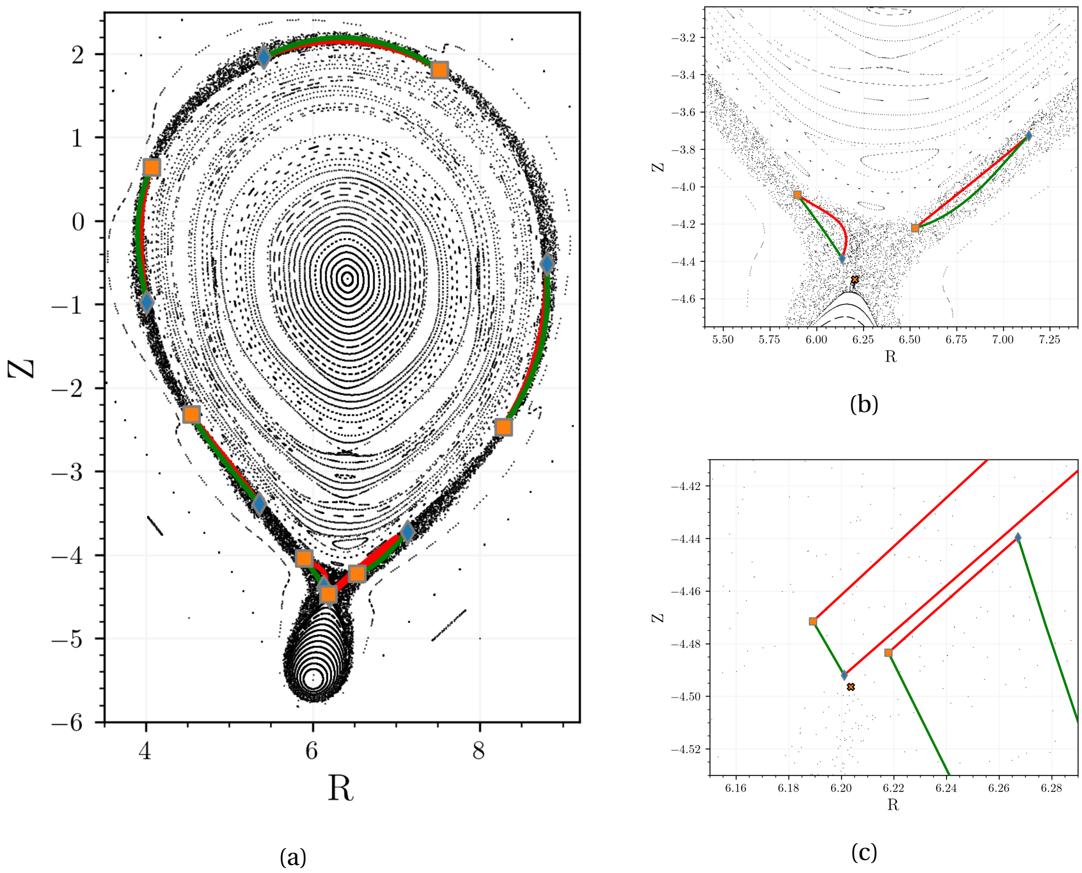


Figure 3.6: Evolution of the turnstile lobe boundary for forward and backward iterations, note here that $\mathcal{P}$ sends points in a counterclockwise manner. The transition to the X-point can be observed in (a) where the manifolds are close to one another. In (b) and (c) the stretching of $\mathcal{P}^k(\gamma_0)$ and $\mathcal{P}^{-k}(\gamma_1)$ can be witnessed as well as the straightening/shrinking of $\mathcal{P}^{-k}(\gamma_0)$ in red and $\mathcal{P}^k(\gamma_1)$ in green.

Fig. 3.6 shows the forward and backward mappings of the contour of a lobe and thus the evolution of the joining curves. The initial lobe is on the low field side : the right lobe in Fig. 3.6b. It is mapped around counter clockwise as shown in Fig. 3.6a. The integral on the homoclinic trajectory (3.5) is shown in Fig. 3.7 and for the five first iterations are performed forward only. It makes a transit from the low field side to the high field side and the value of the integral is jumping around. Then from that point, the last four iterations are forward and backward iterations making the joining curves shrink, as shown in Fig. 3.6b and Fig. 3.6c. Making the integrals converge. After the ninth iteration the points are again in the range where the error due to integration make the regime not linear and the point starts getting further away. However as seen in Fig. 3.6c the algorithm can be stopped before by approximating the joining integrals where W^s and W^u are straight. The calculation of the turnstile flux gives a numerical value of $\Phi_{lobe} = -3.879 \cdot 10^{-14}$.

[4] Again without specifying the unit of the Toybox.

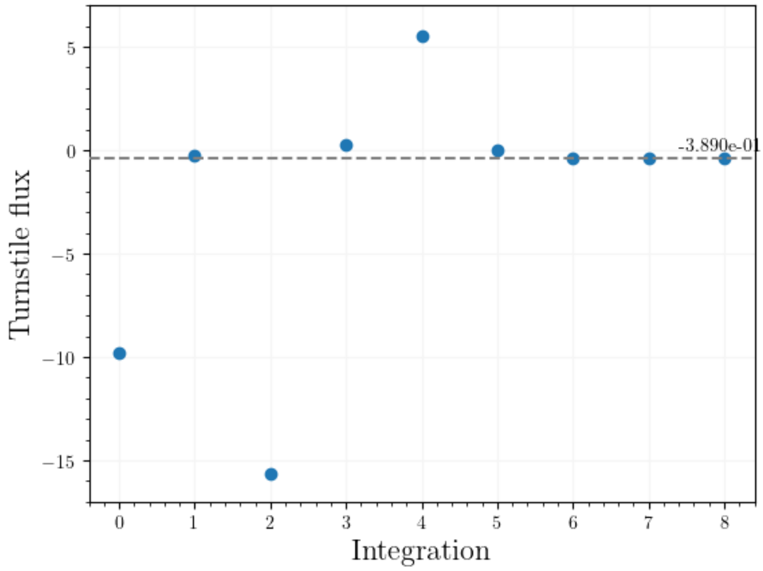
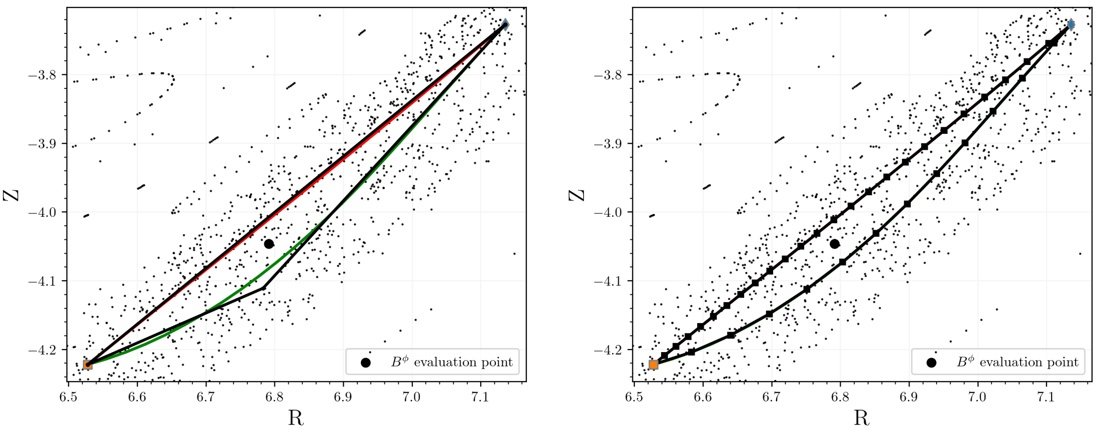


Figure 3.7: Lobe flux Φ_{lobe} calculation as in Eq. 3.5 against the number of iteration of the Poincaré map $\mathcal{P}$. The first five iterations are done forward only. The remaining four iterations, before the integration error causes the algorithm to stop, make the joining integrals small by forward and backward iterations.



(a) (b)

Figure 3.8: Surface and evaluation point for the approximation of the flux through the lobe. In (a) a simple triangle is considered and in (b) the boundary of Σ_{lobe} is found by tracing points in W^s and W^u. The B^{ϕ} component of the magnetic field is evaluated at a random inner point of the lobe.

To verify that the calculated value of the turnstile flux is making sense, Φ_{lobe} can be estimated using the simple approximation that :

$$\Phi_{lobe} = \int_{\Sigma_{lobe}} \mathbf{B} \cdot \mathbf{dS} \approx \tilde{B}^{\phi}_{x_0} \, |\Sigma_{lobe}| \tag{3.6}$$

with $\tilde{B}^{\phi}$ evaluated at the point x_0 inside of the lobe and $|\Sigma_{lobe}|$ the area of the lobe. Choosing the triangle approximation shown in Fig. 3.8a for the surface gives an area $|\Sigma_{lobe}| \simeq 2.9439 \cdot 10^{-2}$. Evaluating $\tilde{B}^{\phi}$ at the point $(R, Z) = (6.791, -4.046)$, shown as the black dot, gives $\tilde{B}^{\phi} \simeq -14.262$ and results in an estimated value for the flux $\Phi_{lobe} \simeq -4.198 \cdot 10^{-1}$, which is close to the calculated value .

This simple approximation of the area can be refined by using the so-called Shoelace formula on the boundary loop Fig. 3.8b computed by tracing field line : setting starting points on W^s and W^u in between h and m. Using the same guess for $\tilde{B}^{\phi}|_x$ and obtaining that $|\Sigma_{lobe}| \simeq 2.728 \cdot 10^{-2}$ gives an estimated flux $\Phi_{lobe} \simeq -3.890 \cdot 10^{-1}$, even closer to the calculated flux. Finally, using the same loop, the integral of $\mathbf{A} \cdot \mathbf{dl}$ can also be approximated using the midpoint rule and determine $\Phi_{lobe} \simeq -3.858 \cdot 10^{-1}$. The redundancy in the value obtained here and the great agreement between them lets us believe that the turnstile flux is correctly calculated.

3.3.2 Impact of the perturbation amplitude on the turnstile flux

Now that we have the calculation of the turnstile flux ready, it will be possible to determine some experimental law for its relationship in some real cases. Here however we first look at the relationship with the amplitude of the perturbation in the $m/n = 6/1$ case in Fig. 3.9. It gives the turnstile flux Φ_{lobe} as a function of the amplitude for applied in the single null Toy-Tokamak. The evolution of Φ_{lobe} is linear with respect to the amplitude with a slope of approximately 3.5. It nicely goes to zero when $\varepsilon_{\text{amp}} = 0$ which is expected, if you do not have a perturbation then there is no turnstile and every point on the separatrix is an homoclinic. For the range scanned from 0 to 0.7 there are no failure in the algorithm. For amplitude bigger we see that the process fails sometimes which is due to the homoclinic search not succeeding. The density of succeeded calculation becomes lower as more as ε_{amp} is increased.

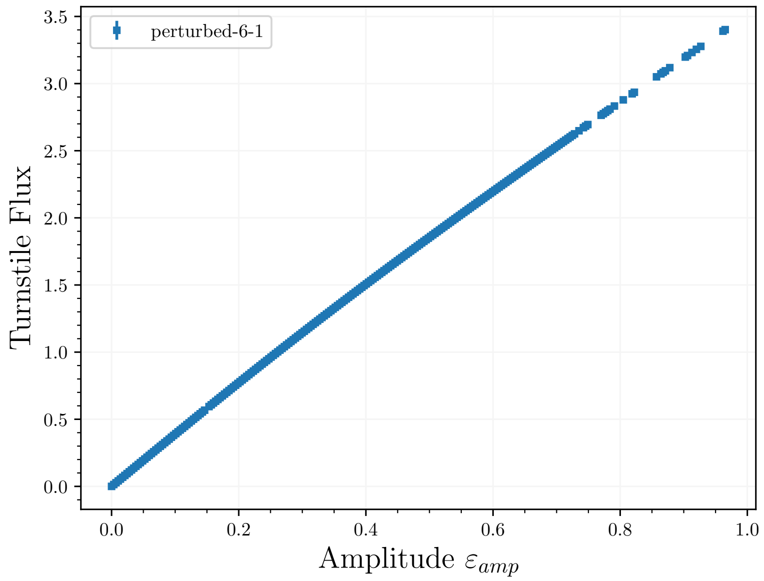
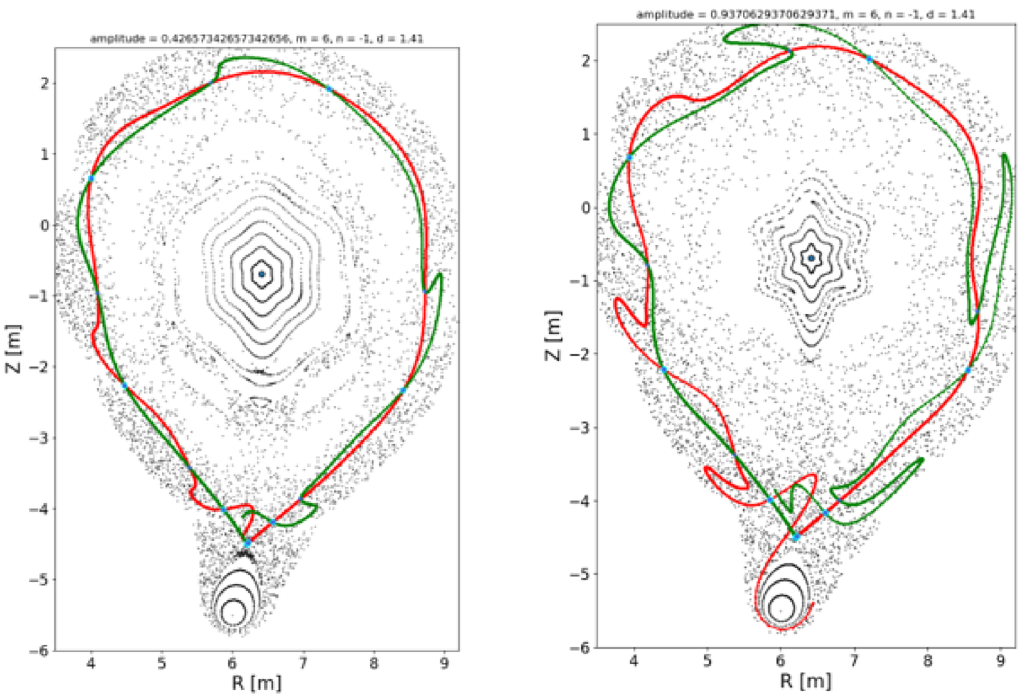


Figure 3.9: Turnstile flux against the amplitude ε_{amp} of the 6/1 perturbation applied on the single null Toy-Tokamak equilibrium.

This was performed be trying to find automatically the two homoclinics. Fig. 3.10 shows the Poincaré section for selected value of the ε_{amp} between 0 and 1 ; it can be seen that the manifold structure are becoming really complicated while the algorithm is still able to find calculate the flux for most of the time. It can be seen how the lobes, in the case $\varepsilon_{\text{amp}} = 0.94$, starts elongating around : the safety factor profile causes the more outward part of the tangle to rotate counter clockwise slower in forward iteration, unstable manifold in red, and inversely for the stable manifold. While it was growing by expanding outward, it is now starting to come back to the manifolds. The background field is nearly only a chaotic sea, and the name ‘perturbation’ looses its meaning.

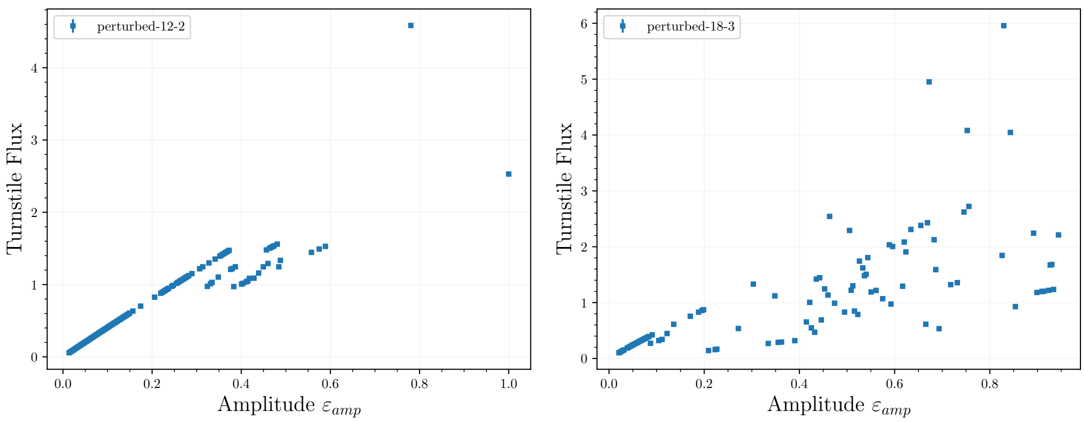
In the case of the 18/3 we have seen that there were 6 different homoclinic orbits to be found and for the 12/2 there are 4. In Fig. 3.11 we see a small linear range compare to the 6/1 case for both higher order perturbation. It is possible that the tangle re-intersects itself changing the number of homoclinics points and causing the search to be wrong.[5]

[5] Maybe a primary-secondary flip or other process from Hohloch (2017).



(a) (b)

Figure 3.10: Poincaré section for the perturbed equilibrium with the manifold plotted at for (a) and (b).

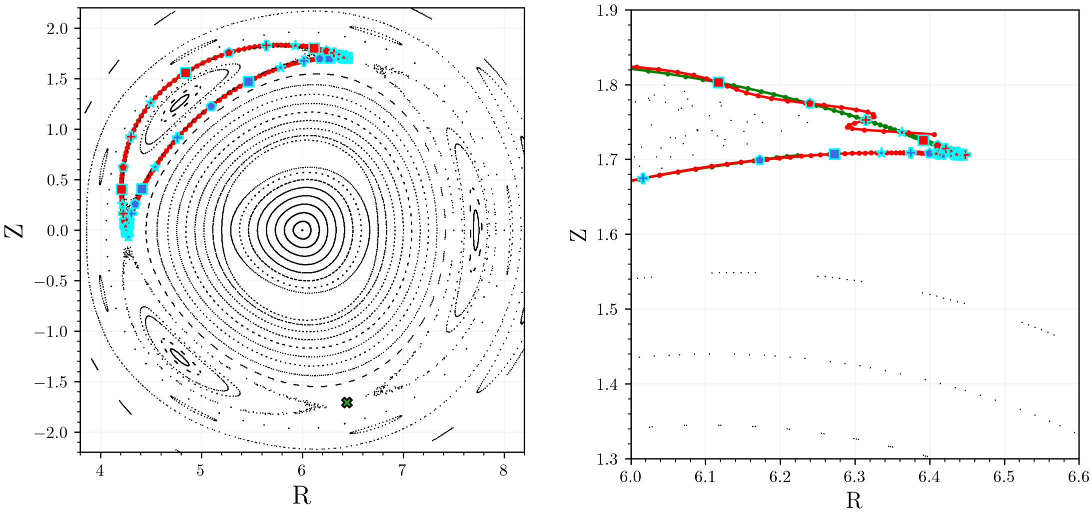


(a) (b)

Figure 3.11: Turnstile flux against amplitude when applying a (a) 12/2 and (b) 18/3 perturbation to the single null Toy-Tokamak.

3.3.3 Turnstiles of a m/n island chain

The turnstile calculation developed in the last sections was discussed in the case of the broken separatrix of a single null Tokamak equilibrium. However the method is not restricted to Tokamaks or $m = 1$ islands, but can be used for any islands/island chain with $\iota = n/m$. The adaptation just requires to use the same amount of iteration as for finding the fixed point, effectively mapping back to the same position in the Poincaré section.



(a) (b)

Figure 3.12: Inner and outer stable and unstable manifolds for the high aspect ratio Tokamak equilibrium with a $m/n = 3/2$ perturbation: (a) general view, (b) zoomed in near a saddle point. The heteroclinics for inner and outer manifolds are shown.

In Fig. 3.12 shows the tangle on the inside 'inner' and on the outside 'outer' tangle. Both the inner (blue) and outer (red) tangles have 4 heteroclinic orbits, the different markers. A closer look at the upper X-point shows how the inner turnstile flux $\Phi_{\text{inner}} = 4.17e-4$ is an order of magnitude lower than the outer turnstile flux $\Phi_{\text{outer}} = 29.47e-4$. There is here 4 different lobe, 2 entering and two exiting lobes. The sum of the lobes is 0. for the inner and 2.3e-5 for the outer one, effectively showing that $\mu(I) = mu(E)$.

The properties of the manifolds of a saddle point were recalled and a general way to compute their shape, as of Wei and Liang (2023), was elaborated. Their intersections were introduced, in particular the primary homo/heteroclinics. An optimisation method and a refined way of setting the initial guess were shown to allow to find the desired intersection in general. However, it was also pointed out that this is the less robust part of the proposed algorithm. An efficient way of finding and knowing how many intersections have to be find remains. This still needs to be investigated.

The main part was that we adapted Meiss's action principle to the case of the field line map. We were able to reduce the integral of the contour and only two homoclinic/heteroclinic trajectories per turnstile that we want to compute. This allowed us to see the effect of increasing the amplitude on the turnstile flux, and a linear relationship was observed. The good behaviour of the algorithm in an island case was finally demonstrated with the introduction of inner and outer turnstile fluxes.

4 Turnstiles in Stellarator fields

The tools introduced in Ch.3 in the case of the *ToyBox* of Ch.2 can be applied equally well to a more general magnetic field of an arbitrary toroidal device. In Stellarators, where both the poloidal and toroidal components of the magnetic field are generated, in equilibrium, by a set of coils and associated currents, the Biot-Savart law can be used to determine the value of the field at any point in space. Therefore, for an existing or hypothetical configuration, the Poincaré map $\mathcal{P}$ can be defined and the turnstile fluxes can be calculated, thus allowing to quantify the stochasticity, the degree of chaos, present due to a given island chain.

Concretely, the framework provided by *Simsopt* from Medasani et al. is employed to calculate the field. It is a very general and well-structured way of dealing with stellarator configurations. It is intended for stellarator design and optimisation and interacts well with some other plasma-specific codes, such as VMEC or SPEC. Simsopt can be used to calculate the field using Biot-Savart directly for each evaluation or to interpolate the field on a mesh, which greatly speeds up the calculation but is less accurate. It also enables the evaluation of the magnetic vector potential $\mathbf{A}$.

The heteroclinic search is the longest part of the turnstile flux calculation process and is conducted using the interpolated $\mathbf{B}$. On the other hand, the direct calculation of $\mathbf{B}$ is used to more precisely search for X-points and determine the turnstile flux.

4.1 QUASR configurations

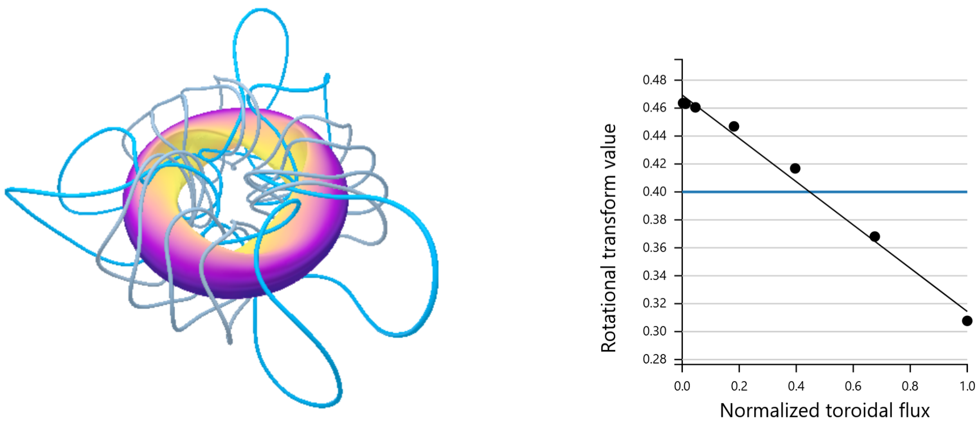
The *QUAsi-symmetric Stellarator Repository*, *QUASR*, contains at the moment over 320'000 stellarator configurations and was created/assembled thanks to Andrew Giuliani.

The search of configurations can be tuned by coil length, number of field periods, type of quasi-symmetry and many other settings. The database explores the space of possible Stellarator fields, which is infinite, and catalogues a number of possible outcomes of Stellarator optimisation.

Three hand-picked configurations with major differences in edge magnetic topology are reviewed in this section. This examination suggests that obtaining turnstile fluxes from edge islands of the hypothetical configurations is a feasible process and their inclusion as a target of optimization for the design of future devices is viable and imminent.

4.1.1 Many islands

The configuration 0229079 is a $n_{\text{fp}} = 3$ quasi-axisymmetric stellarator with 3 coils per field period with many small islands and chaos in the edge. Fig. 4.1a shows the full view of the coil configuration as well as a representative magnetic surface. There are two types of coils with two different value of current. Per field period there is two smaller coils closer to the magnetic surface and one bigger that has a highly curved geometry. The rotational transform profiles ranges from 0.46 at at the axis to a bit more than 0.3 in the edge and the mean value is 0.4.



(a) (b)

Figure 4.1: QUASR configuration 0229079. (a) Coil structure and an external magnetic surface, the same colours indicate the same value of current flowing into the coils. (b) Rotational transform against the normalised toroidal flux and its mean value as a horizontal line.

The Poincaré section for this configuration is shown in Fig. 4.2. It can be seen that there is a lot of intricacy between islands and closed flux surfaces in the edge. The X-points of a $n/m = 6/16 = 0.375$ island chain are found. There is then two periodic orbit of 8 crossings making the total 16 crossings that can be seen in the cross section. Comparing the $\iota\!\!\!-$ profile with the resonance m, n would suggest a value for the normalized toroidal flux to be close to 0.7. The inner most and outer most tangles of the island are shown in Fig. 4.3.

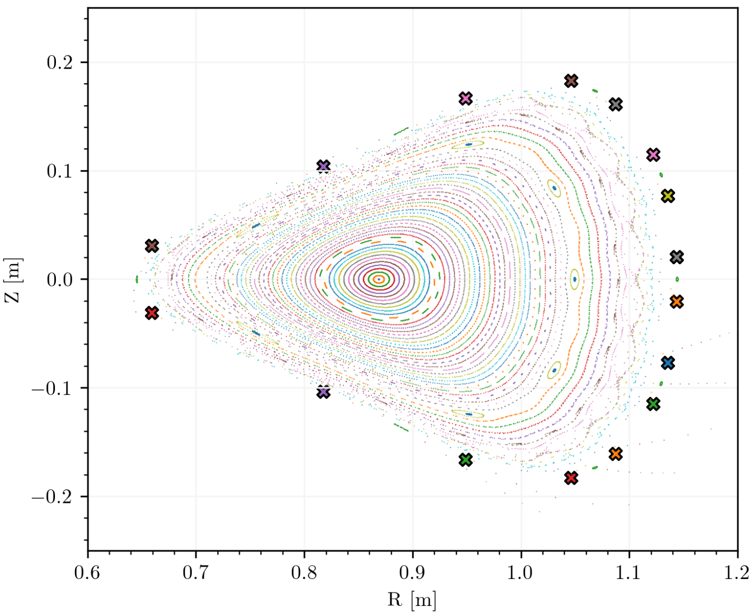


Figure 4.2: Poincaré section at $\phi = 0$ of the QUASR configuration 0229079. THe X fixed points of the island chain $n/m = 6/16$ are drawn as X.

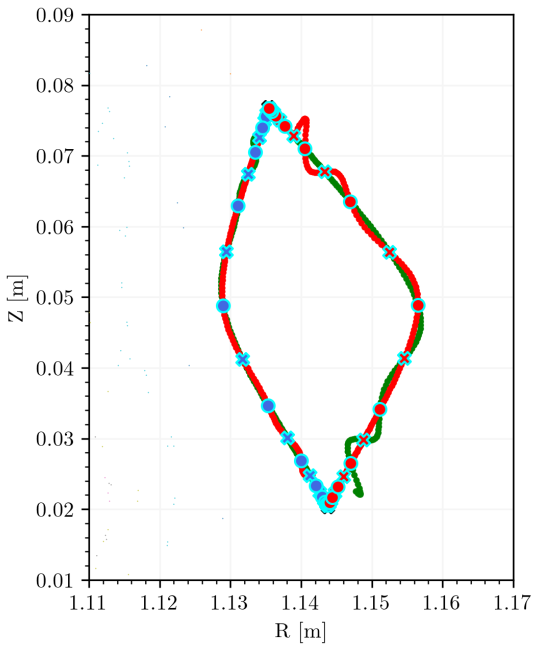
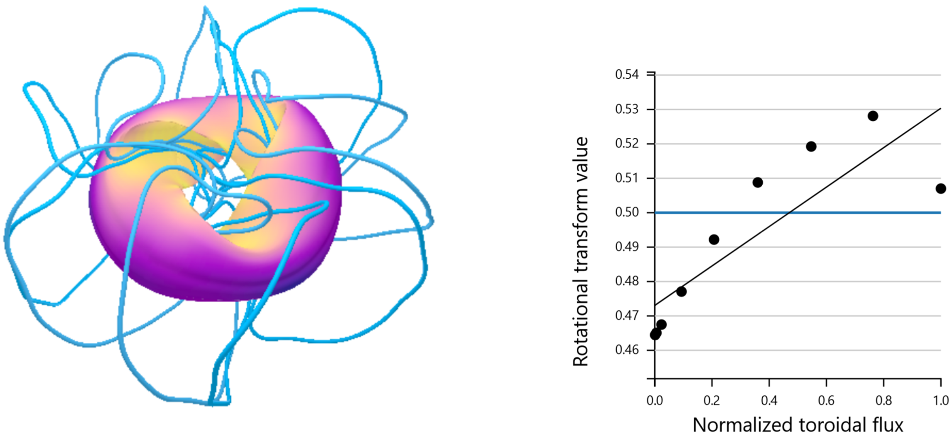


Figure 4.3: Inner and outer tangle and heteroclinic points for an island crossing of the $n/m = 6/16$ chain. There are two heteroclinic orbits for which the crossings are indicated by two different markers. The inner homoclinics are coloured blue, the outer ones red.

4.1.2 Chaotic edge

The configuration 0928241 is a three field period quasi-axisymmetric stellarator. It has two coils per field period and the current through them are equal.Fig. 4.4a shows their geometry and we see that they are long and wrinkled. The rotational transform profile Fig. 4.4b goes up from a value of about 0.46 on the axis to a $\iota\!\!\!-(0.8) \simeq 0.53$ and then goes back down again. The mean value for $\iota\!\!\!-$ is 0.5.



(a) (b)

Figure 4.4: QUASR configuration 0928241. (a) Coil structure and an external magnetic surface, the same colours indicate the same value of current flowing into the coils. (b) Rotational transform against the normalised toroidal flux and its mean value as a horizontal line.

Fig. 4.5 shows the Poincaré plot for the configuration. First we see that there is a $n/m = 6/12$ island chain close to the edge in the inner part and it is very pointy. A $n/m = 6/12$ island chain is also found in the chaotic region and his O and X points are showed here. The island chain is composed of 2 rotating islands. The inner and outer turnstiles are display in Fig. 4.6. The inner and outer turnstiles are non-negligible with respect to the size of the resonance zone.

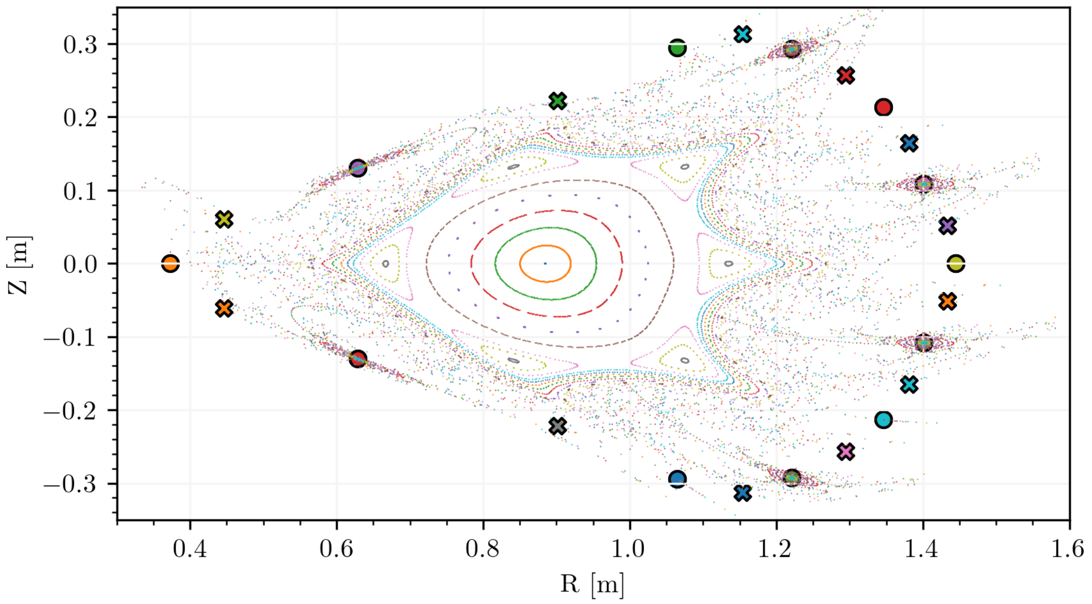


Figure 4.5: Poincaré section at $\phi = 0$ of the QUASR configuration 0928241. The X/O fixed points of the island chain $n/m = 6/12$ are drawn as X and O marker.

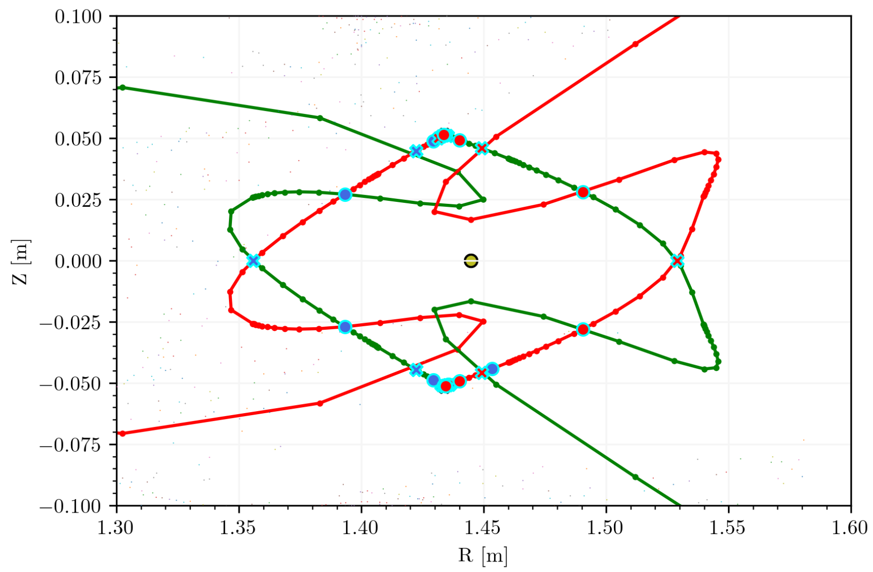


Figure 4.6: Inner and outer tangle and heteroclinic points for an island crossing of the $n/m = 6/12$ chain. There are two heteroclinic orbits for which the crossings are indicated by two different markers. The inner homoclinics are coloured blue, the outer ones red.

The starting point in order to compute the Poincaré plot where set around the two O-point orbit of the 6/12 chain. However while the point around $(R, Z) \approx (1.35, 0.1)$ display closed surface behaviour only few chaotic points were plotted in the resonance zone of the O-point by $Z = 0$. Is it that the island have just different size and that the second one is just really small ?

In fact, writing that O-point are elliptic point is not the best definition. Indeed there is a more topological definition of the difference between O/X-point relying on their topological index. It means that in addition of elliptic points, point for which $\lambda_s, \lambda_u < 0$ are also a different kind of O-point, an alternating hyperbolic fixed point Smiet et al. (2020). Depending on the value of the trace one can differentiate alternating hyperbolic O-point, from elliptic O-point and from hyperbolic X-points.

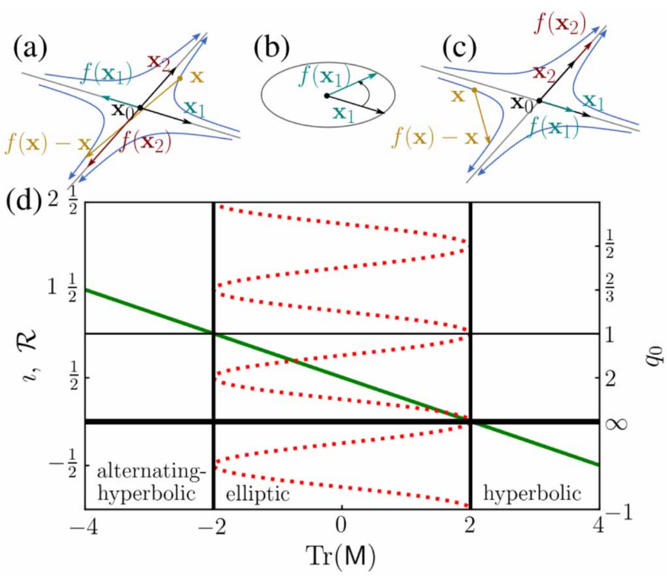


Figure 4.7: Characterization of the fixed point by the value of the trace of the jacobian matrix $\mathcal{DP}^k$ of the field line map (written M here) taken from Smiet et al. (2020).

Tab. 4.1 gives the Greene's Residue $\mathcal{R} = \frac{1}{2} - \frac{1}{4}\mathrm{Tr}(\mathcal{DP}^k|_{x^\star})$, for the 4 fixed points orbits of the 6/12 island chain. Relating the value of the residue to the trace and the condition for hyperbolic $\mathrm{Tr} > 0$ and alternating hyperbolic $\mathrm{Tr} < 0$ fixed point, we see that $R > 1$ fives elliptic point, $R < 0$ gives hyperbolic and $R \in]0,1[$ is the residue of an alternating hyperbolic. Looking back at the table we observed that while the first and second points have the same $\mathcal{R} = \mathcal{R}_1 = \mathcal{R}_2$ and indicating X-points. The O-point where closed surface are seen is an elliptic O-point while the last one is an alternating hyperbolic O-point ![1]

[1] It has in fact tangle structure that takes the form of a Möbius strip, I think.

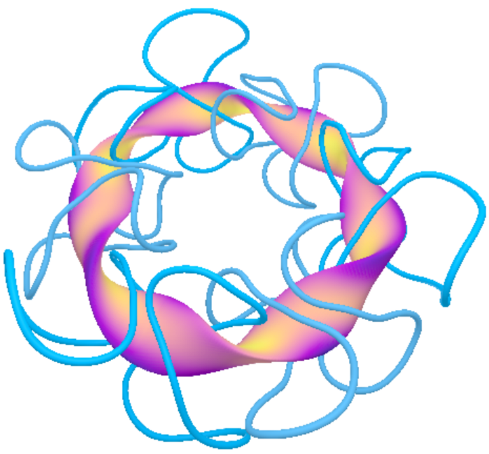
	(1.42, −0.05)	(1.42, 0.05)	(1.35, 0.1)	(1.46, 0)
Residue $\mathcal{R}$	-2.16	-2.16	1.10	0.55

Table 4.1: Greene's residue calculated for each of the four fixed point orbits in the 6/12 island chain, noting their approximating (R, Z). The two X-points have the same value for $\mathcal{R} = \mathcal{R}_1 = \mathcal{R}_2$. The two O-points, however, have different values for their residue.

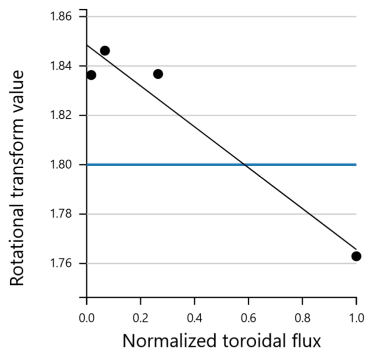
4.1.3 Large island

The last QUASR configuration which will be looked at is the configuration 1329594. In Fig. 4.8, the coil configuration is shown. The coil structure is really interlaced. It seems like the field periodicity is equal to six but in fact it is equal to 3. the number of coil per field period is two and the . it is a quasi-helically symmetric stellarator. Its aspect ratio is of 9.97.

It was looked for high mean $\iota\!\!\!-$ which is the case here at a value of 1.8 and in fact it is surely why this almost looks like a $n_{\text{fp}} = 6$ device. indeed a high iota value mean more twisting, therefore doing the turn taking a $n_{\text{fp}} = 3$ and going to a near doubling gives a higher iota. The iota profile calculation has only four points which is not much and only one in the edge, it goes from about 1.84 to 1.76 at the edge.



(a)



(b)

Figure 4.8: QUASR configuration 1329594. (a) Coil structure and an external magnetic surface, the same colours indicate the same value of current flowing into the coils. (b) Rotational transform against the normalised toroidal flux and its mean value as a horizontal line.

Fig. 4.9a shows the Poincaré section. The island is of the order of a third of the main region and the profile is really elongated. Only one orbit does all the crossings and the island is a $n/m = 3/5$. It is not a clear relation with the iota profile, it may also be that this device in reality a $n_{\text{fp}} = 6$ device for which the optimization was set for a $n_{\text{fp}} = 3$ and which gave back this. This would mean that the $\iota\!\!\!-= 6/5 = 1.2$ which may be more realistic.

Nevertheless the the turnstiles are small compared to the size of the island and the configuration has very little chaos. An optimization could still try to reduce the chaos further by shaping the coils or adjusting the current just a little bit to make this turnstiles, and in particular the outer one smaller. Or to maybe to combine a somewhat big island with a bigger turnstile to. The question is whether or not we want chaos here, for example, but we can now make it more or less chaotic.

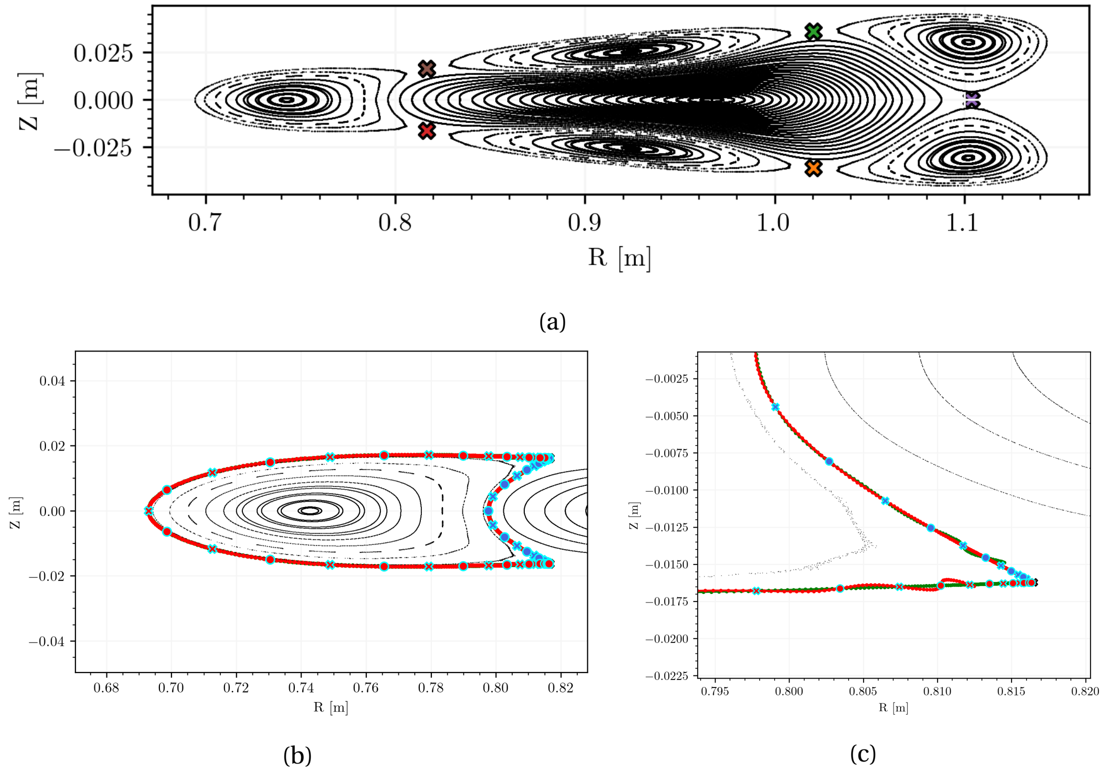
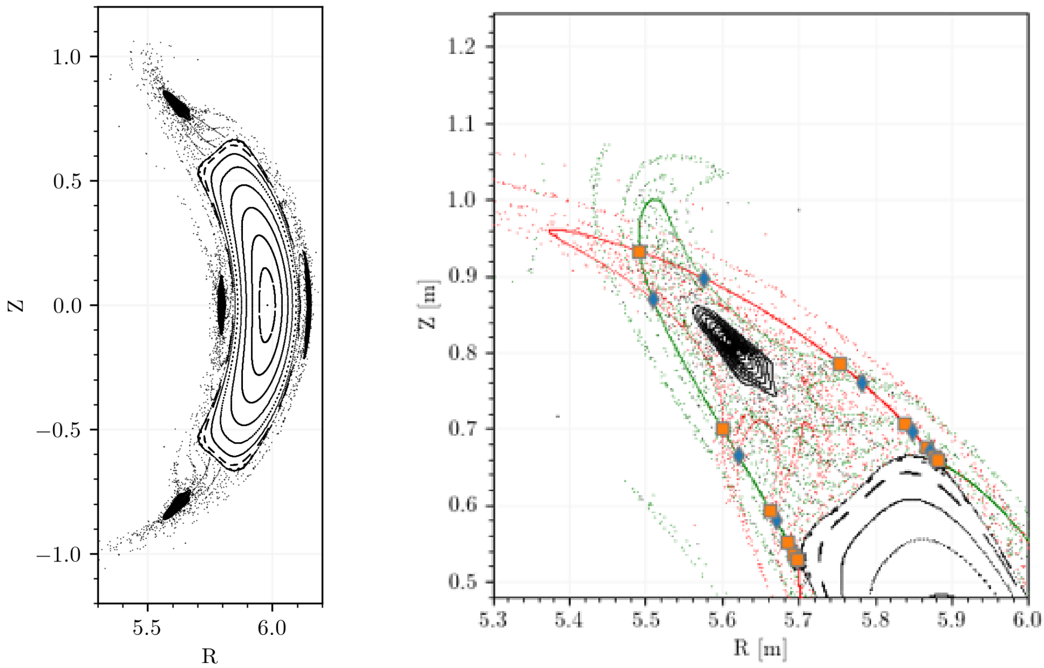


Figure 4.9: Poincaré section at $\phi = 0$ of the QUASR configuration 1329594 in (a) with the X-points of the of island $n/m = 3/5$ drawn as X marker. In (b) : look at the manifold structure around the left island crossing. In (c) : closer around the lower X-point of the island. The two heteroclinic orbits are indicated by distinct markers : circles and crosses. The inner ones are coloured in blue, the outer ones in red.

4.2 Wendelstein 7-X

Wendelstein 7-X is a 5 field period device with a total of 50 modular coils located in Greifswald, Germany. Per field period, in addition to the 10 modular coils that generate the standard W7X field, 4 planar coils can be used to further vary the magnetic field. In this existing stellarator experiment, the shape of the coils is difficult, if not impossible, to alter. The current, on the other hand, can be easily adjusted to vary the magnetic topology.

The standard configuration, shown in Fig. 1.1a, is obtained by running a current of 1.62 MA in the modular coils and none in the planar coils. This configuration of W7X was designed to be non-chaotic and exploits the 5/5 island chain in the edge as an island divertor concept. However, other more chaotic configurations can be explored. By reducing the currents in the modular coils to 1.1095 MA and those in the planar coils are set to -0.3661 MA creates the GYM00-1750 obtained by Robert Davies. Its Poincaré section Fig. 4.10a displays a smaller $n/m = 5/4$ island and a chaotic edge region.



(a)

(b)

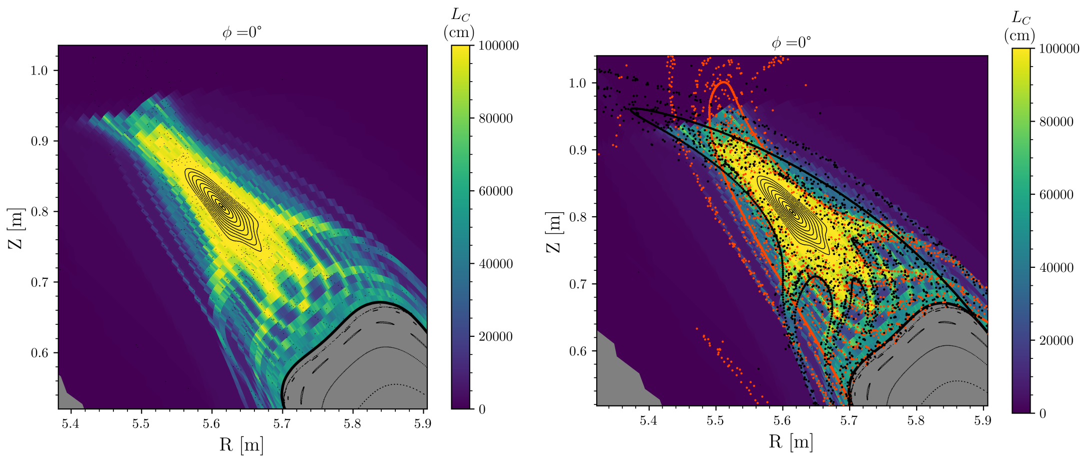
Figure 4.10: Wendelstein 7-X chaotic GYM00-1750 configuration Poincaré plot (a) and tangle structure with heteroclinics in (b). A $n/m = 4/5$ island can be observed and chaos is present in the edge.

The tangle structure created by the stable and unstable manifold is shown in Fig. 4.10b. The infinite lattice intersects many times and the primary heteroclinics are nicely located. In this context, in order to trace W^s and W^u with higher precision, the field is evaluated directly from the Biot-Savart law and the integration is carried out, using Runge-Kutta Dopri5, with a relative tolerance of 1e-14. However, after a small time evolution, the initial segment still becomes a fuzzy layer and not a perfect 2d line. Tuning the integrator to the device or perhaps using a different integration scheme may help reduce the fuzziness and make the tangles clearer. Although not to be trusted blindly, the Runge-Kutta algorithm is very robust for many ODE systems and is relied upon here.

A field line on a closed surface will remain inside of the magnetic island forever. A field line from the chaotic layer, on the other hand, will eventually leave the resonance zone as it gets into the exiting set of the turnstile. Then, when on the outside, it will end up hitting the plasma-facing components. The connection length L_C Fig. 4.11a shows this difference in the behaviour of the field lines. The distance travelled before hitting the PFC, i.e. before hitting the computational boundary surface that terminates the integration, is displayed. Higher values therefore mean a longer confinement time and are indicated by a more yellow colour.

An interesting structure emerges by looking around the closed surface of the island. The connection length is high in certain local regions and then low close by. Several layers of connection length intensities can also be distinguished. Putting in relation 4.10b and Fig. 4.11a gives Fig. 4.11b where it can be observed that the tangle structure is identical, and in fact defines, the different connection length zones. The fact that the stable and unstable manifolds are the boundary of these zones brings us back to thinking about the definition of exit and entry sets. Remember that a point in the exit set will be mapped out of the resonance zone while a point in the entry set will be mapped in. As both forward and backward integration are included there are in fact two entering and two exiting sets. If I_f and E_f are the entry and exit sets of the forward evolution, then $I_b = \mathcal{P}(E_f)$ and $E_b = \mathcal{P}(I_f)$ are the entry and exit set of the backward evolution.

The resonance zone has therefore 2 exit sets when considering both evolution. A point in the entry set of the forward evolution will in fact have a minimum in the backward evolution and similarly for points in the entry set of the backward evolution which should give low values of L_C has the field line is already outside. When inside the minimum is related to the actual length, number of iteration before being in either E_f or E_b. While the last argument is not rigorous, the link between the tangle structure and the connection length is striking, which certainly implies a relationship with the entry and exit sets.



(a) (b)

Figure 4.11: Minimum connection length between backward and forward tracing for the chaotic W7-X configuration, computed using EMC3-Lite. In (a) around the top island crossing and in (b) overlaying the computation of the manifolds of the saddle.

4.3 Inner and outer turnstile fluxes

The turnstile fluxes have been calculated for the tangles of the specific island considered in each of the stellarator configurations presented. For magnetic islands, there is a tangle closest to the magnetic axis, the inner tangle, and a tangle farthest from the axis, the exterior or outer tangle. There were only 2 heteroclinics for both tangles in every configurations look at. Tab. 4.2 gives the flux calculated for the inner and outer turnstiles for each configuration, normalised by the toroidal component of the magnetic field evaluated at the axis $\tilde{B}_0^{\phi}$. The value of the inner turnstile fluxes are, for all the magnetic fields analysed here, lower than for the outer turnstiles. In the case of the chaotic W7X configuration it is even two order of magnitude bigger on the outside than on the inside.

	$\Phi_{inner}/\tilde{B}_0^{\phi}$ $[10^{-5}]$	$\Phi_{outer}/\tilde{B}_0^{\phi}$ $[10^{-5}]$
QUASR 0229079	1.66	2.29
QUASR 0928241	46.5	135.8
QUASR 1329594	0.14	3.77
W7X GYM00-1750	0.82	63.00

Table 4.2: Inner and outer turnstile fluxes normalized by the ϕ component of the magnetic field at the axis for the different stellarator configurations. The three QUASR configurations discussed and the W7X chaotic configuration.

An intuitive argument about the implication on the stochastic transport is that it is bigger on the outside than on the inside. This difference can be intuitively seen as if the whole island now acts like a custom officer. To go from the inner region to the inside of the island requires to enter by the inner turnstile gate which is small. However once inside the second one can be passed much easily as it is a much bigger turnstile.

In this last section we have applied the method developed in Ch.3 to the case of a magnetic field given by a set of coils and currents, more specifically in stellarators. Three different configurations were considered in terms of the topology they present in their boundary region. A chaotic configuration of W7X was also analysed, showing the interesting relationship between tangle/turnstiles and connection length when chaos is present. Finally, the higher value of the turnstile flux further away from the axis with respect to the inner turnstiles was a constant throughout the islands observed.

5 Conclusion

The chaotic behaviour of the magnetic field in magnetic fusion devices at the edge has a major impact on the impact pattern on the plasma-facing components. In particular, they need to be studied for the future design of the plasma-facing divertors, which will have to withstand high thermal loads.

A simple definition was given for the field line map : which takes an initial point in a given toroidal section and maps it to the next field period. The fixed point of that map were shown to be closed field lines and O/X points, depending on the eigenvalues of the Jacobian.

An analytical toybox has been introduced to simplify the calculation and to be able to test the further algorithm while still using the field line tracing paradigm used for real cases. The toybox requires only the analytical formulation of the vector potentials and computes the resulting fields by automatic differentiation. Its basic block is closed nested toroidal flux surfaces with a toroidal component that generates a quadratic safety factor profile. Other potentials can be added to this building block, including the axisymmetric potential of a circular current loop and, importantly, perturbations. Both were used to make a single null analytical Tokamak equilibirum. The perturbation added in toroidal coordinates, non-straight field, quickly induces chaos and two distributions have been implemented for the radial profiles, Maxwell-Boltzmann and Gaussian.

Applying a 6/1 Maxwell-Boltzmann perturbation to the single null equilibrium, we were first able to find its x-point. Then the algorithms : to compute the stable and unstable manifold of this hyperbolic fixed point. to find the homoclinic intersections were presented. It was found that while the 6/1 had two homoclinic points, 12/2 and 18/3 for example had 4 and 6 different homoclinic orbits.

Using an adaptation of Meiss's work in the case of the standard map, we were able to demonstrate a method of calculating the turnstile flux and show that the resulting numerical value was indeed meaningful. This flux is at the core of quantifying the degree of chaos due to a certain island or around the Tokamak separatrix. The algorithm was used to show the linear dependence of the turnstile flux with increasing perturbation amplitude for a wide range of values starting from zero.

Then, using the coils, currents of specific stellarator configurations and the Biot-Savart law, we were able to calculate the turnstile fluxes in the case of 3 hypothetical QUASR stellarators. It was shown that the calculation can be performed on realistic stellarator coil, and directly using the biot-savart of a coil set and that the turnstile flux can be used for future optimisation of stellarator designs. In the case of W7X, we showed how the connection length plot of a chaotic edge configuration relates to the tangle structure created by the stable and unstable manifold of the 4/5 island present.

Finally, for those stellarator islands, the fact that there is an inner and outer tangle structure with a distinct value for the turnstile flux was discussed, which will certainly have an impact on transport, and it was argued that it will act as a sort of customs office for the line coming from the inside. However, a more profound argument is certainly desirable.

With the additions presented in this work, pyoculus can be used to study the transport due to chaos and have not only an eye, but a microscope into chaos. The main aspect where the method presented in this document should be improved is in finding the homo/heteroclinic intersections of manifolds. Quantifying the number of intersections is an interesting question, the answer to which will hopefully reveal many more questions.

A Geometry concepts

The space $\mathcal{M} := \mathbb{R}^3$ with the usual euclidean inner product is seen in differential geometry as a Riemannian manifold. Roughly defined, for each point $p \in \mathcal{M}$, the space of linear differential operator at p is a tangent plane that passes through p noted $T_p\mathcal{M}$. The usual way we think about vector may now be to think about $v = \sum_i v^i \frac{\partial}{\partial x^i} = v^i \partial_i \in T_p\mathcal{M}$, with the Einstein summation convention. Then a vector field is at each point p a vector in the tangent space at p. The inner product g_p can also be defined for each point and then for the two vector field $u, v \in T\mathcal{M}$ the metric g is its value at each point. For a choice of coordinates (chart) g can be expressed explicitly and if we choose the Cartesian (x, y, z), Cylindrical (R, ϕ, Z) and Toroidal (ρ, ϕ, θ) coordinates, then it can be written as :

$$
\begin{aligned}
g &= dx \otimes dx + dy \otimes dy + dz \otimes dz, \\
g &= dR \otimes dR + R^2\, d\phi \otimes d\phi + dZ \otimes dZ, \\
g &= d\rho \otimes d\rho + (R_a + \rho \cos(\theta))^2\, d\phi \otimes d\phi + \rho^2\, d\theta \otimes d\theta.
\end{aligned}
$$

Either one of the expression for g can be transform into the other by a change of basis $x \leftrightarrow y$ with the rules that dx changes in a contravariant way and ∂_x as a covariant way ;

$$
\begin{aligned}
\frac{\partial}{\partial y^{i'}} &= \frac{\partial x^i}{\partial y^{i'}} \frac{\partial}{\partial x^i} \\
dy^{i'} &= \frac{\partial y^{i'}}{\partial x^i} dx^i.
\end{aligned}
$$

For instance if we take x the cartesian coordinates and y the cylindrical coordinates :

$$
\begin{aligned}
g = dx \otimes dx + dy \otimes dy + dz \otimes dz &= \delta_{ij}\, dx^i \otimes dx^j \\
&= \delta_{ij} \frac{\partial x^i}{\partial y^{i'}} \frac{\partial x^j}{\partial y^{j'}} dy^{i'} \otimes dy^{j'} = \frac{\partial x^i}{\partial y^{i'}} \frac{\partial x^i}{\partial y^{j'}} dy^{i'} \otimes dy^{j'} \\
&= dR \otimes dR + R^2\, d\phi \otimes d\phi + dZ \otimes dZ.
\end{aligned}
$$

To bother introducing those definition has a really elegant advantage. If we note more the fact that there exists a way to go from $T_p\mathcal{M}$ to the space of dx and vice and versa by the lowering $\flat$ and raising $\sharp$ operators. And introduce another operator called the Hodge star dual $\star$, then the usual differential operators on a function f or on a vector field F become :

$$\begin{aligned}
\nabla f &= (df)^\sharp \\
\nabla \cdot F &= \star d \star (F^\flat) \\
\nabla \times F &= (\star d(F^\flat))^\sharp \\
\Delta f &= \star d \star df
\end{aligned}$$

which are expressions without explicit appearance of the coordinates! The tricky way into the geometry is now worth the journey. For example, the gradient in cylindrical coordinates :

$$\nabla f = (df)^\sharp = (\partial_R f dR + \partial_\phi f d\phi + \partial_Z f dZ)^\sharp = (g^{R,R}\partial_R f)\partial_R + (g^{\phi,\phi}\partial_\phi f)\partial_\phi + (g^{Z,Z}\partial_Z f)\partial_Z$$

with the element of the inverse of the metric g. Defining the orthonormal basis as $e_R = \partial_R$, $e_\phi = 1/R\partial_\phi$ and $e_Z = \partial_Z$ we see that it gives the expected gradient :

$$\nabla f = \partial_R f \partial_R + \frac{1}{R^2}\partial_\phi f \partial_\phi + \partial_Z f \partial_Z = \partial_R f\, e_R + \frac{1}{R}\partial_\phi f\, e_\phi + \partial_R f\, e_Z.$$

A.1 Integration of the action

For the algorithm in Sec.3.3.1, we need to integrate the vector potential $\mathbf{A} \in T\mathbb{R}^3$ is $\mathbf{A} = A^i \partial_i^{\text{cyl}}$ along the trajectory of the homo/hetero-clinics. In cylindrical coordinates the curvilinear integral of the scalar product $\mathbf{A}\cdot\mathbf{dl}$ along γ between 0 and ϕ is :

$$\int_0^\phi g(\mathbf{A}, \mathbf{dl}) = \int_0^\phi g(\mathbf{A}(\gamma(s)), \dot\gamma(s))ds = \int_0^\phi (A^R\dot\gamma^R + R^2 A^\phi \dot\gamma^\phi + A^Z\dot\gamma^Z)ds$$

For example, integrating along a toroidal loop of constant radius R in the axisymmetric case gives $R^2 A^\phi = 2\pi\psi$ which links ψ with the poloidal flux ψ_p.

A.2 Flux conservation with differential forms

The field line map defined in Eq. 1.1 is flow preserving. It was shown in section 1 that it implies the relation Eq. 1.3 for the determinant of the Jacobian matrix $\mathcal{DP}$. A simpler way is to use differential forms. If we write the flux in the form formalism, then $\beta = B^\phi dR \wedge dZ$ and the integral becomes :

$$\iint_\Sigma \mathbf{B}\cdot\mathbf{dS} = \iint_{\mathcal{P}(\Sigma)} \mathbf{B}\cdot\mathbf{dS} \Leftrightarrow \int_\Sigma \beta = \int_{\mathcal{P}(\Sigma)} \beta = \int_\Sigma \mathcal{P}^\star\beta$$

with $\mathcal{P}^\star\beta$ the pullback of β through the field line map $\mathcal{P}$. Intuitively it takes the form which was in $\mathcal{P}(\Sigma)$ back to Σ. Then the flux conservation becomes $\mathcal{P}^\star\beta(\mathcal{P}(x)) = \beta(x)$ and using the relation between the differential forms :

$$\begin{aligned}\mathcal{P}^\star\beta &= \beta_{i'j'}\, d(\mathcal{D}\mathcal{P}^{i'}_{\ i} x^i) \wedge d(\mathcal{D}\mathcal{P}^{j'}_{\ j} x^j) \\ &= \beta_{i'j'}\left(\mathcal{D}\mathcal{P}^{i'}_{\ i}\mathcal{D}\mathcal{P}^{j'}_{\ j} - \mathcal{D}\mathcal{P}^{i'}_{\ j}\mathcal{D}\mathcal{P}^{j'}_{\ i}\right) dx^i \wedge dx^j = \beta_{ij}\det(\mathcal{D}\mathcal{P})\, dx^i \wedge dx^j\end{aligned}$$

and we see that it implies here :

$$\det(\mathcal{D}\mathcal{P}) = \beta_{RZ}(x)/\beta_{RZ}(\mathcal{P}(x)) = B^\phi(x)/B^\phi(\mathcal{P}(x))$$

and we got the same formula back as a direct calculation.This shows the power of differential forms.

B Implementations

B.1 Potential

Using Mathematica (Inc) to integrate the toroidal field $-B^{\phi}$ from Eq. 2.2 by dZ gives the A^R vector potential as :

$$
\begin{aligned}
A^R = \frac{1}{4r}\mathrm{Real}[&\left(4 \cdot q_a + \alpha_s \cdot \left(5r^2 - 10rR + 4R^2 + 2(z-Z)^2\right)\right) \cdot \sqrt{-r^2 + 2rR - (z-Z)^2} \cdot (z-Z) \\
&- ir(r-2R) \cdot \left(4 \cdot q_a + \left(3r^2 - 6rR + 4R^2\right) \cdot \alpha_s\right) \cdot \log\left(-iz + \sqrt{-r(r-2R) - (z-Z)^2} + iZ\right)]
\end{aligned}
$$

which is implement as it is using the Numpy library from JAX.

B.2 Jacobian of $\mathcal{P}$ in details

The Jacobian of the field line map as a matrix form $\mathcal{DP} := \partial \mathcal{P}^{\{\tilde{R},\tilde{Z}\}} / \partial\{\tilde{R}, \tilde{Z}\} \in \mathbb{R}^{2\times 2}$. Here we distinguish between R, Z in the starting plane, noted as $\tilde{R}, \tilde{Z}$, and the general evolution around the torus $R = \gamma^R, Z = \gamma^Z$, which is a handy abuse of notation. For instance :

$$
\mathcal{DP}^{\tilde{R}}_{\tilde{R}} = \frac{\partial}{\partial \tilde{R}} \left[\int_{\phi_i}^{T+\phi_i} \dot{\gamma}^R(\phi) d\phi\right] + 1 = \int_{\phi_i}^{T+\phi_i} \partial_{\tilde{R}} \dot{\gamma}^R(\phi) d\phi + 1 = \int_{\phi_i}^{\phi_i+T} \partial_{\tilde{R}} \left[\frac{B^R}{B^{\phi}}\right] d\phi + 1 = \ldots
$$

with B^R and B^{ϕ} being evaluated at :

$$
\begin{aligned}
B^R &= B^R(R(\tilde{R}, \phi, \tilde{Z}), \phi, Z(\tilde{R}, \phi, \tilde{Z})) \\
B^{\phi} &= B^{\phi}(R(\tilde{R}, \phi, \tilde{Z}), \phi, Z(\tilde{R}, \phi, \tilde{Z})).
\end{aligned}
$$

The integrand can then be developed using the chain rule :

$$\begin{aligned}
\partial_{\tilde{R}}\left[\frac{B^R}{B^\phi}\right] &= \partial_R\left[\frac{B^R}{B^\phi}\right]\partial_{\tilde{R}}R + \partial_Z\left[\frac{B^R}{B^\phi}\right]\partial_{\tilde{R}}Z = \frac{1}{B^\phi}\frac{\partial B^R}{\partial \tilde{R}} - \frac{B^R}{(B^\phi)^2}\frac{\partial B^\phi}{\partial \tilde{R}} \\
&= \frac{1}{B^\phi}\left(\frac{\partial B^R}{\partial R}\frac{\partial R}{\partial \tilde{R}} + \frac{\partial B^R}{\partial Z}\frac{\partial Z}{\partial \tilde{R}}\right) - \frac{B^R}{(B^\phi)^2}\left(\frac{\partial B^\phi}{\partial R}\frac{\partial R}{\partial \tilde{R}} + \frac{\partial B^\phi}{\partial Z}\frac{\partial Z}{\partial \tilde{R}}\right).
\end{aligned}$$

Without showing the same kind of equality for the other integrands, we can write in matrix form that :

$$\mathcal{DP} = \int_{\phi_i}^{T+\phi_i} \begin{pmatrix} \partial_R\left[B^R/B^\phi\right] & \partial_Z\left[B^R/B^\phi\right] \\ \partial_R\left[B^Z/B^\phi\right] & \partial_Z\left[B^Z/B^\phi\right] \end{pmatrix} \cdot \begin{pmatrix} \partial_{\tilde{R}}R & \partial_{\tilde{Z}}R \\ \partial_{\tilde{R}}Z & \partial_{\tilde{Z}}Z \end{pmatrix} d\phi + \mathbb{I}_2$$

B.3 Pyoculus Newton

In order to find the fixed point of the field line map applied k times $\mathcal{P}^k$, the Newton method is used :

$$x_{n+1} = x_n - J_F(x_n)F(x_n)$$

with $F(x) = \mathcal{P}^k(x) - x$ and J_F its jacobian in matrix form. Two ideas can be used to find the fixed point of a m/n island chain. We can first simply try to find the fixed point of the map applied k times, where k is the order of the orbit of the fixed point. This is what is being done in most cases in this paper. However, we may find point that do not have the right winding number around. For instance the magnetic axis is always a fixed point for any number of application of $\mathcal{P}$. Another way is to pass to the poloidal coordinates centered at the axis for the RZ plane and to record the evolution of the angle. Writing H the change from poloidal to cartesian in the plane and using the chain rule gives the needed F and J_F quantities :

$$\begin{aligned}
F(\rho,\theta) &= \mathcal{P}^k(H(\rho,\theta)) - H(\rho,\theta) \\
J_F(\rho,\theta) &= J_{\mathcal{P}^k}(H(\rho,\theta))J_H(\rho,\theta) - J_H(\rho,\theta).
\end{aligned}$$

The condition that $\Delta\theta(k) = 2\pi$ can be used to be sure that the right $\iota\!\!\!-$ is found. Both methods were found to work, but the first seems to converge a bit more. The second works best when you are trying to optimise for only one coordinate, for example when you know that a fixed point is at z=0 using stellarator symmetry.

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